MRO Reverse Engineering Study

Final Report

Team 13

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1 Introduction

Mars Reconnaissance Orbiter (MRO) is a multipurpose spacecraft launched on August 12, 2005. The mission is still ongoing, providing important information to get a better knowledge of morphology and climate of Mars and helping guide other missions on and around the planet. The following paragraphs present an overview of the mission and its technical aspects.

1.1 Mission High-Level Goals

The primary objectives of the MRO mission are to [113]:

- Acquire information about the present climate of Mars and its seasonal behavior.
- Identify water-related geographic formations and searching for sites showing evidence of hydrothermal activity.
- Characterize in detail the stratigraphy, geomorphology, and composition of the Mars surface and subsurface at many globally distributed locales to improve understanding of the nature of different types of Martian terrain and their changes over time.
- Identify sites with the highest potential for future Mars missions.
- Transmit and receive scientific data from spacecrafts (S/Cs) involved in Mars missions during after the science goals are achieved.
- Test and provide data to aid in the development of experimental engineering payloads.

1.2 Mission Drivers

In the MRO mission the following drivers, which have strongly influenced the design of the S/C, can be identified:

- **Data handling** The high-resolution data acquisition performed by the payloads (P/Ls) require a high-performance data processing and transmission system to deliver the data efficiently. The MRO was set out to collect and transmit an unprecedented amount of data for an interplanetary mission. For the Primary Science Phase (PSP) alone, 34 terabits of data was scheduled to downlink, which is more than three times the amount five comparable, previously launched, interplanetary missions together had transmitted [38]. A three meter diameter parabolic antenna was necessary to downlink said data in such amounts. As such, data handling is a key driver of the mission, with significant downstream effects on the rest of the mission's design.
- **Aerobraking** In order to achieve mission success, aerobraking was considered the most suitable option to inject into the Primary Science Orbit (PSO), due in large part to the fuel —and thereby mass—saved (aerobraking reduces the amount of fuel that must be launched from Earth by nearly one-half). Although the Mars Global Surveyor and Mars Odyssey previously utilized aerobraking [66], the maneuver ended up bending a solar panel of the former, therefore making the MRO mission an attempt to improve the braking mechanism on trajectory to Mars. The decision to utilize aerobraking must be made early in the S/C design process, as it significantly changes the desired properties and configuration of the S/C components, particularly those of the solar panels and antenna. Other important aspects, influenced in their totality by aerobraking, are the structural and thermal stresses to which the S/C is subject during this phase. Finally, being a complex and high-risk maneuver, it required continuous attitude and orbital corrections, hence introducing new challenges to the design.

1.3 Functional Analysis

The MRO mission can be broken down into the functional blocks in Figure 1.

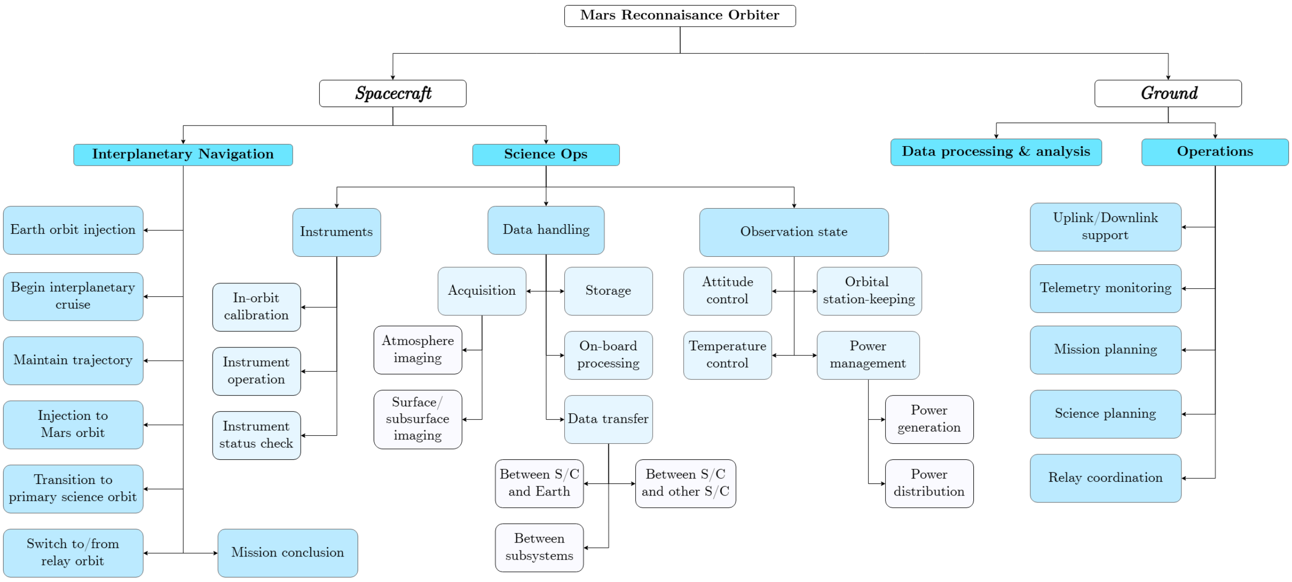


Figure 1: Functional analysis diagram of MRO.

1.4 Mission Phases

The operational mission of the MRO is divided into several key phases [113]:

1. **Launch and Early Orbit Phase (LEOP)**: MRO was launched aboard an Atlas V-401 rocket. After a brief period of Earth coasting, Centaur upper-stage injected it on an interplanetary transfer trajectory.

2. **Cruise and Approach Phase (CAP)**: After launch MRO began its seven-month journey to Mars. MRO followed a Type-I ballistic trajectory to deliver the orbiter on a southern approach trajectory to Mars. The seven-month cruise phase was partitioned into two distinct mission stages. The first one included spacecraft navigation, and checkouts, while the second one focused on Mars approach.

3. **Mars Orbit Insertion (MOI)**: MRO executed a propulsive maneuver to enter Mars orbit using its main engines while adjusting attitude with smaller thrusters. This phase ended with spacecraft reconfiguration and imaging operations in order to prepare for the next phase.

4. **Aerobraking and Transition Phase (ATP)**: During this phase, the spacecraft dropped into the upper portions of the Martian atmosphere and used the aerodynamic drag to stabilize spacecraft into a near-circular 2-hour orbit. Transition was designed to deliver MRO with a series of maneuvers into PSO. The MRO spacecraft then underwent a series of final checkouts prior to commencing primary science operations.

5. **PSP**: The spacecraft began its PSP, during which it conducted high-resolution imaging, atmospheric studies, and mineralogical mapping. The MRO is capable of performing off-nadir rolls of up to 30° for targeted observations and executes routine orbit maneuvers to maintain precision. Onboard logic uses the latest ephemerides and an Integrated Target List (ITL), i.e. a list of targets on Mars on which to perform either sequential or concurrent observations.

6. **Relay Phase (RP)**: The MRO has served as a crucial relay asset for multiple Mars missions, including Phoenix, Curiosity, Perseverance, and InSight. It has assisted during Entry, Descent, and Landing (EDL) by recording telemetry data via its Ultra High Frequency (UHF) antenna, Electra Proximity Link Payload (Electra), and performed orbital maneuvers to optimize coverage. MRO has functioned as a key communication relay, ensuring reliable data transmission between landers and Earth.

7. **Extended Mission**: After completing PSP and RP, the MRO entered the Extended Mission Phase. During this time, it continues to monitor the Martian surface, atmosphere, and climate, and assist future missions by acting as a communication relay.

1.5 ConOps and Functionalities

The chart below demonstrates clearer understanding of the ConOps and their relationships with the mission phases [43, 113]:

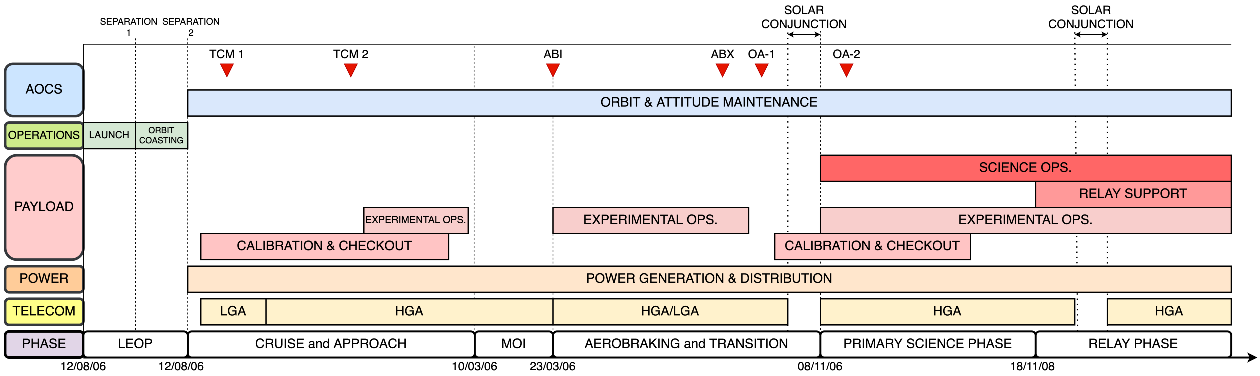


Figure 2: ConOps and phases of the MRO mission (not to scale).

1.6 Mission/System Level Requirements

The main requirements [34] at mission level can be grouped into:

- **Science Operations and Planning**: The orbiter shall operate in the primary science orbit for one Mars year, utilizing all six science P/Ls in targeting, survey, and mapping modes. It shall also conduct gravity and accelerometer investigations. The system shall support simultaneous operations and data acquisition by all science instruments. Additionally, a target acquisition selection process shall be created to ensure conflict-free targeting priorities.

- **Data Acquisition, Transmission and Storage**: The mission shall return a total science data volume of 34 terabits or more, over the one-Mars-year primary science phase, to achieve full mission success, with a minimum success criterion of 26 terabits [113]. At least 95% of acquired data shall be provided to spacecraft operators and science investigators. The system shall have the capability to properly manipulate and analyze data coming from scientific investigations and the Electra relay payload. Science downlink data rates shall support up to 6 Mb/sec, and the spacecraft shall provide onboard storage for at least 100 Gb of science-related data.

- **Navigation and Orbital Control**: The orbiter's position and velocity relative to Mars shall be predicted with an accuracy of 1.5 km in the downtrack direction, 0.05 km in the cross-track direction, and 0.04 km in the radial direction. The Navigation System shall ensure that, after

MOI , the periapsis altitude remains within 200-400 km for at least 8 consecutive orbits without additional maneuvers. The spacecraft shall be capable of pointing anywhere within a ±30° cone about the nadir direction in the PSO. The orbit shall be near circular, at a low altitude, and near polar to allow for detailed imaging, complete coverage of the Mars surface, and a repetition of possible targets every 17 days.

- **Structural, Thermal and Power**: The spacecraft shall incorporate thermal protection and dissipation solutions to withstand the increases in temperature caused by aerobraking and the heat generated by multiple electric P/Ls and subsystems operating simultaneously. The structural design shall be able to withstand the torques and forces experienced during launch and aerobraking, especially on the joints of the appendixes [76]. The power system shall be able to supply the required electrical power to all P/Ls and instruments operating in parallel.
- **Mission Lifetime**: The mission shall ensure a minimum operational lifetime of one Mars year for the science payloads. The telecommunication instruments, including Electra and the High-Gain Antenna (HGA), should remain operational until 2009. Additionally, the spacecraft shall carry enough fuel to support operations until 2010 [113].

1.7 Payload

1.7.1 Scientific Instruments Overview

The instruments on the MRO include six main scientific payloads, as well as communication systems to support the mission's objectives of studying the Martian surface, atmosphere, and subsurface, while ensuring high rates of data downlink, uplink, and on-board handling [67].

- **HIgh Resolution Imaging Science Experiment (HiRISE)**: It is a high-resolution nadir pointing telescopic camera with a primary mirror diameter of 50 cm and a 1.15° Field-of-view (FOV) [66]. It allows to obtain images of a vast area of the planet while being able to see details on the surface. Near-infrared observations provide an unprecedented view of layered materials, channels and geological formations in general. Among other duties, HiRISE also allows to identify appropriate landing sites for future Mars missions [67].
- **Compact Reconnaissance Imaging Spectrometer for Mars (CRISM)**: This spectrometer has a telescope of 10 cm with a 2° FOV and a maximal resolution of 18 m/pixel over a swath of 11 km by using its articulated mechanism to slow the apparent speed of image on the ground [66]. The CRISM uses infrared and visible spectrometers to detect certain minerals on the surface that indicate water has been present. It also acquires data for atmospheric survey.
- **Mars Climate Sounder (MCS)**: It is a spectrometer made up of two telescopes, each with a 4 cm aperture, mounted on a yoke frame that can move to autonomously adjust its pointing direction [66]. The instrument can measure temperature, humidity and dust content, and observe the atmosphere both visible and infrared light. The profiles generated are useful to predict Martian weather and climate, as well as improve understanding of the variations of the polar caps.
- **Mars Color Imager (MARCI)**: It consists of two framing cameras, one with two spectral channels in the UV band and the other with five channels in the visible band. MARCI produces a global weather map of the planet to obtain a view of daily, seasonal and year variations as well as atmospheric events like dust storms.

- **Context Camera (CTX)**: It is composed by a mono-chromatic camera working in a single wavelength band for visible light. It is mainly used to provide a background view of the terrain around smaller targets for interpreting the high-resolution observations captured by other instruments, such as HiRISE or CRISM .
- **SHAllow Subsurface RADar (SHARAD)**: It is a nadir-looking synthetic aperture radar sounder designed to probe beneath the Martian surface to a depth of up to 1 km. SHARAD plays a key role in the research of water traces, scanning the subsurface in search of strong radar returns which suggest the presence of underground liquid or frozen water with a resolution of between 0.3 km and 3 km and the vertical resolution is approximately 15 m.
- **Additional Payload**: The mission includes a relay telecommunications package and two technology demonstrations [113]. The Ka-band system compares Ka and X-band, showing that Ka-band requires less power to transmit the same amount of data [67]. Electra, a UHF package, improves the landing precision of landers, aids the location of the rover, and provides a relay on the Mars network [67]. The Optical Navigation Camera (ONC) tracks Mars moons, refines moon ephemerides, and supports precise entry trajectories for landers [66].

Figure 3 shows the placement of the P/L in the spacecraft:

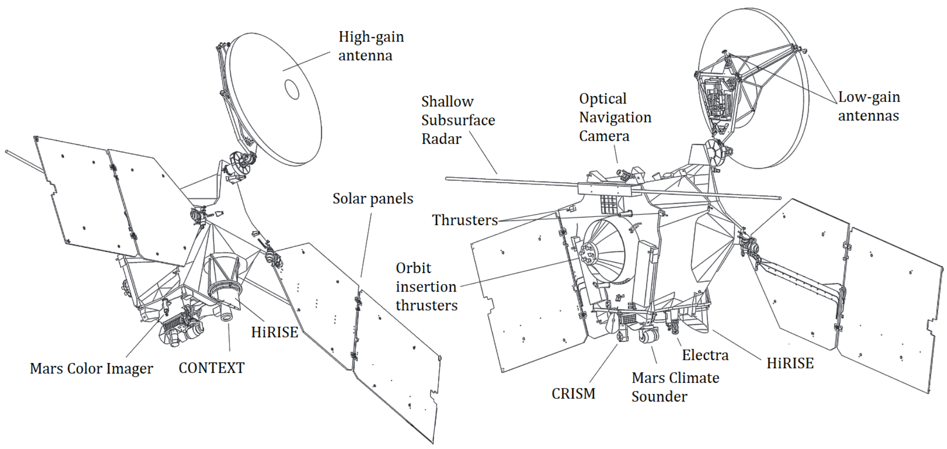


Figure 3: Components of Mars Reconnaissance Orbiter [66]

1.7.2 Additional Science Experiments

During its lifetime, the spacecraft itself (without additional payloads) was used to study the atmosphere and the gravitational field of Mars. In the aerobraking phase, accelerometer data was used to determine atmospheric density at different altitudes. The Doppler shift in the radio communication signal was used to study the gravitational field of Mars.

1.8 Payload - Phases, functions and objectives correlation

Table 1 highlights the correlation between the payload functions, the mission objectives listed in section 1.1, and the phases in which they have been involved. It can be noticed that some P/Ls have been used in multiple phases to achieve goals different from their main one. Apart from calibration and testing operations, which were primarily scheduled during the cruise phase and just before the PSP, imaging P/Ls were used to properly prepare for the aerobraking phase, while the MCS was used to retrieve information about the atmosphere during aerobraking.

Payload	Payload Functions	Mission Objectives	Phase
HiRISE	High Resolution Near-IR & Visible Imaging	To study active surface processes and landscape evolution	MOI , PSP, RP
CRISM	Spectrometric Surface & Atmospheric Analysis	To search for the residue of minerals that form in the presence of water	PSP, RP
MCS	Atmospheric Observation by Composition and Temperature	To observe the temperature, humidity, and dust content of the Martian atmosphere, making measurements that show variations in Mars' current weather and climate	ATP, PSP, RP
MARCI	Surface & Weather Mapping	To produce a global weather map of Mars to help characterize daily, seasonal, and year-to-year variations in the red planet's climate	MOI , PSP, RP
CTX	Wide & Context Imaging	To help provide a context for high-resolution analysis of key spots on Mars provided by HiRISE and CRISM	MOI , PSP, RP
SHARAD	Scanning of Mars Subsurface Composition	To look for indications of liquid or frozen water up to 1 km into Mars' crust	PSP, RP
Ka-band System	High-Rate Data Transmission Experiment	To provide information to improve performance in deep space communications	PSP, RP
Electra	Functions as a Node in the Mars Network	To provide relay communications for crafts on and around Mars	RP
ONC	Optical Tracking	To improve navigation capability, to identify microsatellites in Mars equatorial plane, to improve landing accuracy of rovers and landers	CAP, PSP, RP

Table 1: The payload functions and mission objectives of MRO.

1.9 Mission Analysis

For the mission profile selected, the launch period spanned four weeks, beginning from 10 August 2005, with a minimum launch window of 90 minutes each day. Of the initial four trajectory correction maneuvers (TCMs) and two contingency maneuvers (to be executed in the event of a Mars-impact trajectory), ultimately, only two TCMs were required to reach target orbit at Mars. After aerobraking, the MRO was inserted into a low-altitude (250 X 315 km), near-circular, near-polar (92.65°) and frozen orbit with its periapsis fixed over the south pole. The choice of a nearly polar orbit ensures complete coverage of Mars over many orbits, while the altitude and near-circularity allow to obtain an acceptable trade-off between low drag effects and high-quality imaging. The choice of altitude was chosen to yield a repeating ground track such that nearly all locations on the planet can be viewed within 17 days by pointing the spacecraft off-nadir by up to 20° [113].

2 Propulsion Subsystem and Mission Analysis

2.1 Mission Trajectory Design

In the next section, mission trajectory design is analyzed, dividing the flight into phases and reporting the ΔV costs for each maneuver.

2.1.1 Low Earth Operations and Cruise Phases

During the cruise phase, MRO followed a Type-I ballistic trajectory between Earth and Mars, with a change in true anomaly less than 180° between its departure (12/08/2005) and arrival (10/03/2006) dates. The interplanetary trajectory met these constraints [32]:

- 21-day launch window for high success probability.
- Excess velocity under 3 km/s for cost-effective orbital capture.
- Arrival date and nodal orientation suitable for aerobraking into the Primary Science Orbit.

Atlas V was selected for its efficiency with a high Declination of Launch Asymptote (DLA) ($\geq 41°$) and high injection energies (C3 $\geq 20\,\mathrm{km^2/s^2}$)[32]. Five TCMs were planned based on covariance analyses predicting navigation uncertainties. TCM-1 corrected the upper stage injection errors of the launcher, while TCM-2 addressed the execution errors of TCM-1 and the uncertainties of residual orbit determination. TCM-3 and TCM-4 were planned as drift mitigation. A contingency maneuver (TCM-5) was included to prevent an unexpectedly low orbit altitude [110].

2.1.2 Mars Orbit Insertion and Aerobraking

On 10/03/2006, the spacecraft began the MOI maneuver, designed to match the hyperbolic trajectory from launch and cruise with the starting conditions of aerobraking. The maneuver placed MRO into an high-elliptical orbit with apoapsis at 44 982 km, periapsis at 426 km, and a period of 35.5 h. On 30/03/2006, aerobraking began, aimed at lowering the apoapsis to 486 km. This maneuver exploits the effects of atmospheric drag, allowing to save up to 1200 m/s in ΔV. During the aerobraking various maneuvers were planned [32, 110]:

- **Corridor Control Aerobraking Maneuver (ABM)**: Aims at raising and lowering the periapsis. 26 of these were performed, with a total planned $\Delta V = 33\,\mathrm{m/s}$. In particular, the first six of them lowered the periapsis from 426 km to 108 km during the walk-in, the initial stage of aerobraking.
- **Pop-up maneuver**: Used to raise periapsis to 160 km in case of unsafe anomalies, with $\Delta V = 20\,\mathrm{m/s}$;
- **Aerobraking Termination Maneuver (ABX)**: Performed during the walk-out phase with $\Delta V = 25\,\mathrm{m/s}$, raising periapsis from 111 km to 215 km, marking the end of the aerobraking phase.

2.1.3 Transition and Primary Science Orbit

After aerobraking, MRO executed a series of Orbit Adjustments (OAs) maneuvers to reach the PSO. OA-1 raised periapsis and adjusted inclination, while OA-2 lowered apoapsis and rotated the apse-line from an argument of pericenter of 187° to 270° [110]. The PSO was designed to [32]:

- Enable low-altitude observations.
- Maintain a repeating ground track for rapid mapping, short-term revisit opportunities, and long-term track spacing under 5 km at the equator.
- Minimize atmospheric drag effects.
- Be sun-synchronous for repeatable lighting conditions.

The selected orbit had a Sun-synchronous ascending node at 3 P.M. LMST, a near-polar inclination of 92.7°, and an eccentricity that maintained a low-altitude frozen orbit (255 km periapsis, 320 km apoapsis, 270° argument of periapsis) [110]. Its 3775 km semi-major axis ensured a 17-day ground track repeat cycle. In this phase, Orbit Trim Maneuvers (OTMs) are planned to maintain the sun-synchronous orbit and the repeating ground track. In addition, a ΔV budget equal to 19 m/s has been allocated to station keeping, and the same quantity for the station keeping of the relay phase [32].

2.2 Mission Maneuver Sizing

This section examines mission trajectory design, calculating and justifying ΔV costs through maneuver analysis and station-keeping. Using patched conics method and an impulsive maneuver assumption, maneuvers are assessed with their respective ΔV. Computations of TCM-1 and TCM-2 are implemented based on the propagation of the uncertainty of launcher injection [107] and the uncertainty of TCM-1, respectively [32]. Due to their stochastic nature, 100 % margin has been applied to their costs. For MOI, velocity at the entry of the sphere of influence is determined via a Lambert-based algorithm. The estimated, not margin-adjusted, ΔV for insertion, computed as a maneuver at pericenter, is lower than the real one, due to non-impulsive corrections. Available data allowed to estimate accurately the first six ABMs. These, aimed at lowering the periapsis to 108 km, were modeled as apoapsis maneuvers. During aerobraking, the saved ΔV is estimated via energy balance between the initial and final orbits. The estimated result of saved cost (ΔV = 1994 m/s), deviates greatly from the real one (ΔV = 1182 m/s). This discrepancy is due to the rough approximation given by the energy approach, which does not take into account many properties of the aerobraking maneuver. A more precise atmospheric drag model would give a more reliable estimation. OA-1 was modeled as a plane change, while OA-2 as an apse-line rotation. Deviations are mainly inked to environmental perturbations.

2.3 ΔV Budget Breakdown

Margins follow industry standards [100], based on whether the maneuver is stochastic or deterministic.

Maneuver	**Real ΔV** [m/s]	**Estimated ΔV** [m/s]	**Margins**	**Total ΔV** [m/s]
Cruise Phase				
TCM-1	7.80	15.7	100%	31.4
TCM-2	0.754	0.129	100%	0.258
MOI	1000.5	952.9	5%	1000.5
ABM				
ABM-1	-	4.1	5%	4.3
ABM-2	-	4.2	5%	4.4
ABM (3-6)	-	6.0	5%	6.3
Total	14.5	14.3	5%	15.0
ABX	25.0	26.2	5%	27.5
Transition to Science Orbit				
OA-1	15.0	18.5	5%	19.4
OA-2	53.5	50.6	5%	53.1
Station Keeping	38.0	45.4	100%	90.8

Table 2: Mission maneuver ΔV budget.

2.4 Propulsion Subsystem

The Propulsion Subsystem (PS) of MRO is required to perform orbit maneuvers of varying magnitudes, trajectory correction maneuvers, attitude control, station keeping, and momentum desaturation of the reaction wheels. The primary drivers that significantly influenced the design and architecture of the propulsion system are:

- The mission's reliance on highly critical maneuvers.
- The high frequency of maneuvers over time, particularly during the aerobraking phase.

To meet these requirements, MRO is equipped with a total of 20 monopropellant hydrazine thrusters of three different sizes. The propulsion system operates with blow-down pressurization for all thruster activities except for the MOI burn, where a pressure regulator is employed to ensure consistent thruster performance during this risky maneuver [111].

2.4.1 Propulsion System Architecture and Rationale

Thrusters
To ensure greater robustness, hence being more tolerant to single-fault thruster failures, a configuration with multiple smaller engines is preferred over a single, more powerful thruster [46]. The need to deliver impulses of different magnitudes and in different directions throughout the mission led to the choice of using three distinct types of thrusters as detailed in Table 3.

- The primary engine thrusters (MR-107N) are specifically designed for the MOI burn where a large amount of ΔV should be quickly delivered, firing simultaneously all the 6 engines [43]. To mitigate the risk associated with first-time usage during MOI phase, also TCM-1 is executed using the MR-107N [43, 46]. These thrusters are aligned along the principal axis of the spacecraft.
- The intermediate thrusters (MR-106E) are mainly designed for the TCMs during the cruise phase but they are also used for orbital parameter adjustments during and after the aerobraking phase, thrust vector control during MOI and station keeping maneuvers during the PSP and Relay phases [43, 46]. Due to their dual-role functionality, these thrusters serve both as primary and secondary propulsion. These thrusters are strategically distributed around the spacecraft, oriented to provide thrust in multiple directions, thereby generating both translational and rotational accelerations.
- The smallest thrusters (MR-103D) are primarily utilized for attitude control and for executing angular momentum desaturation (AMD) of the reaction wheels. Additionally, they assist in maintaining the correct aerobraking drag attitude. They are arranged in couples and are fired in pairs to allow differential firing, so that ideally the resulting net ΔV is zero [43, 46].

Thruster	**Number**	**Thrust** [N]	$I_{\mathbf{sp}}$ [s]	P_{cc} [bar]	**Mass** [kg]	**Propellant**
MR-107N (M)	6	170.0	232-229	11.2-4.2	0.74	Hydrazine
MR-106E (T)	6	22.0	235-229	12.4-4.5	0.52	Hydrazine
MR-103D (A)	8	0.9	224-209	23.4-5.9	0.33	Hydrazine

Table 3: Engines' main characteristics.

The masses seen in Table 3 also account for the valve integrated with each system. All the thrusters selected for this mission are flight-proven, with significant heritage from previous space missions. Their extensive operational history provides a solid foundation to meet the propulsive drivers; this knowledge minimizes development risk, enhancing overall mission reliability and predictability, especially during critical mission conditions.

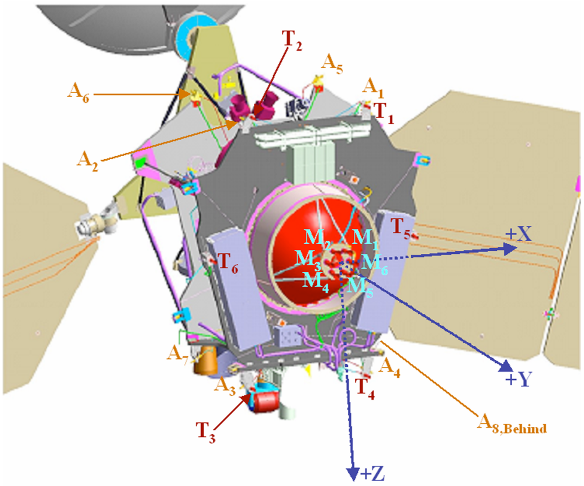


Figure 4: Thrusters and tank configuration [111]

Propellant

The propulsion design drivers required a system capable of doing frequent start-stop operations, of having a throttleable thrust, and characterized by simplicity, reliability, and predictable behavior. A bi-propellant system ensures a higher performance and better efficiency but involves more complexity in architectures. For these reasons, a monopropellant architecture was selected as the most suitable option [41].

Hydrazine is selected over other possible monopropellants mainly due to its extensive heritage in space applications. Its advantages include excellent long-term stability, ease of decomposition within a catalytic bed, and relatively high specific impulse with respect to other monopropellants. However, employing hydrazine introduces specific risks: it is highly corrosive, meaning that engine plumes or potential leaks could compromise critical spacecraft instruments or components.

Pressurant and Feeding Strategy

The pressurant chosen for this mission is helium. It is often utilized in space application because of its low density, its properties, which are consistent with temperature changes, and its non-reactivity with hydrazine. The propulsion system operates in blow-down for most maneuvers, except for the MOI, where a pressure regulator is introduced to guarantee consistent and accurate thruster performance but also to limit the significant fuel consumption of this maneuver. This choice derives from a trade-off between tank dimensions and required precision: a fully regulated system with a larger helium tank, sufficient to maintain constant pressure throughout the entire mission, would have resulted in significant mass and volume penalties. However, relying entirely on a blow-down system would lack the precision and consistency essential for the critical MOI maneuver[43, 48].

Tanks and Pipes

The MRO spacecraft is equipped with a cylindrical propellant tank designed to contain approximately 1187 kg of hydrazine, approximately 70% of which is allocated to the MOI [2][3]. A separate cylindrical tank, positioned on the side of the main propellant tank, contains high-pressure helium gas that, through a regulator, goes into the propellant tank where it puts the hydrazine under pressure. During the cruise and Mars approach phases, the gradual hydrazine consumption led to a gradual pressure decrease in the tank, dropping from a peak of 205 psi to roughly 190 psi following MOI [111]. From the pressurized tank, hydrazine is directed through metal tubing to each thruster that, by a valve, can be fired independently [48].

2.4.2 Reverse Sizing of the Propulsion Subsystem

The reverse sizing of the MRO propulsion subsystem is carried out based on the total ΔV required for the mission, reserve excluded, the dry mass at launch and the engine parameters. These data are taken from the literature and summarized in the Table 4 (refer to Table 3 for thrusters data). Some assumptions have also been made to perform the analysis and they will be underlined in the following paragraphs. During the reverse sizing process, the contributions of all three propulsive systems are considered. To this end, the total ΔV has been divided into three components to consider the different amounts spent by MR-107N, MR-106E and MR-103D engines.

ΔV_{MR107N}	ΔV_{MR106E} [m/s]	ΔV_{MR103D} [m/s]	M_{dry} [kg]	$P_{\text{cc, max}}$ [Pa]	$P_{\textbf{cc, min}}$ [Pa]
1010	385	150	1031	2340000	420000

Table 4: Reverse sizing data of the PS [43].

- **Propellant Sizing** Firstly, a 20% system-level margin is applied to dry mass at launch. The estimated wet mass can be retrieved using the specific impulse of the engines, the ΔV and the standard gravitational acceleration constant $g_0 = 9.81\,\text{m/s}^2$ using Tsiolkovsky's equation:

$$M_{\text{wet}} = M_{\text{dry}}\, e^{\frac{\Delta V}{I_{\text{sp}}\, g_0}}$$

  The estimated propellant mass is obtained by subtracting the dry mass from the wet mass. A 5.5% margin is then applied to this value (3% for fuel residuals, 2% for ullage uncertainty, and 0.5% for loading uncertainty). The volume occupied by the propellant can be computed dividing the obtained mass by the density of hydrazine ($\rho_{\text{prop}} = 1010\,\text{kg/m}^3$). An additional 10% margin is then applied to account for possible unusable volume. The procedure described is applied to all three engine groups, considering the ΔV and specific impulse associated with each. The individual contributions are then summed to obtain the total quantities.

- **Pressurant Sizing** As anticipated in the previous sections, the pressurization system operates in two different ways depending on the mission phases: pressure-regulated and blow-down. The two systems require different reverse engineering approaches, as described below. It is important to note that, due to a lack of information, the initial and final pressures in the gas tank are estimated as follows:

$$P_{\text{gas, f/i}} = P_{\text{cc, min/max}} + \Delta P_{\text{feed}} + \Delta P_{\text{inj}}$$

  where P_{cc} is taken from the engine datasheet, while ΔP_{feed} and ΔP_{inj} represent respectively the losses due to the feeding lines (35 kPa) and the injection system (estimated at 30% of the chamber pressure).

Blow-Down. The blow-down ratio B, computed as the ratio between $P_{\text{gas,i}}$ and $P_{\text{gas,f}}$, allows the calculation of the initial gas volume as follows:

$$V_{\text{gas, bd}} = \frac{V_{\text{prop, bd}}}{B-1}$$

Subsequently, the mass of gas involved in the blow-down system can be computed as:

$$M_{\text{gas, bd}} = \frac{P_{\text{gas, i}} \cdot V_{\text{gas, i}}}{R_{\text{He}} \cdot T_{\text{tank}}}$$

where $R_{\text{He}} = 2077.3\,\text{J}/(\text{kg}\cdot\text{K})$ and the tank temperature is assumed as $T_{\text{tank}} = 293\,\text{K}$. A mass margin of 20% is then applied to the obtained value. This procedure is carried out for both MR-106E and MR-103D engine groups, since these are the systems that primarily rely on a blow-down system.

Pressure-Regulated. During MOI, the pressurization system switches to a pressure-regulated system that maintains the tank pressure constant. The mass of helium required for this phase can be calculated assuming an adiabatic transformation:

$$M_{\text{gas, pr}} = \frac{P_{\text{gas, f}} \cdot V_{\text{prop, pr}} \cdot \gamma_{\text{gas}}}{R_{\text{He}} \cdot T_{\text{tank}} \cdot \left(1 + \frac{P_{\text{gas, f}}}{P_{\text{gas, i}}}\right)}$$

where $\gamma_{\text{He}} = 1.67$ and again $T_{\text{tank}} = 293\,\text{K}$. Another 20% mass margin is applied to this value. The fraction of volume occupied by helium can then be computed using the ideal gas law.

In analogy with the propellant sizing, the different contributions are summed to obtain the total values of mass and volume related to the pressurant.

- **Tanks Sizing** As mentioned before, the propellant and gas tanks are assumed to be cylindrical with two hemispherical caps and made from titanium ($\rho_{\text{Ti}} = 2087\,\text{kg/m}^3$, $\sigma_{\text{Ti}} = 950\,\text{MPa}$). With these reasonable hypotheses, it is possible to size two separate tanks for propellant and pressurant. Firstly, a 1% bladder margin is applied to the tank volume. Holding the assumption of $V_{\text{cyl}} = V_{\text{sph}}/3$, the radius of the tank and the height of its cylindrical part can be computed from geometric considerations.

  Subsequently, the thickness of the spherical part and the cylindrical one can also be obtained as:

  $$t_{\text{sph}} = \frac{P_{\text{gas, i}} \cdot r_{\text{tank}}}{2\cdot\sigma} \qquad t_{\text{cyl}} = \frac{P_{\text{gas, i}} \cdot r_{\text{tank}}}{\sigma}$$

  Having the needed quantities, it is now possible to retrieve the masses of the spherical and cylindrical parts of the tanks:

  $$M_{\text{sph}} = \rho_{\text{Ti}} \cdot \frac{4}{3}\pi \left((r_{\text{tank}} + t_{\text{sph, gas}})^3 - r_{\text{tank}}^3\right)$$

  $$M_{\text{cyl}} = \rho_{\text{Ti}} \cdot \pi \left((r_{\text{tank}} + t_{\text{cyl}})^2 - r_{\text{tank}}^2\right) \cdot h_{\text{cyl}}$$

  The final mass of the tanks can be finally obtained summing the two contributions.

- **Mass Budget** The total mass of the propulsion subsystem (propellant excluded) can be calculated summing the masses of the thrusters (provided in Table 3), the tanks and the gas.

- **Power Budget** The table below shows the maximum power required by each model considering the valve power, the valve heater power and the catalytic bed power of one thruster only per type.

	MR-107N	MR-106E	MR-103D
Power [W]	59.4	49.15	13.72

Table 5: Power budget of the thrusters.

2.4.3 Results Analysis

Table 6 shows a comparison of the estimated data with the real ones available from the mission:

	w/o Margins	**Margins**	**w/ Margins**	**Real**
$M_{\text{propellant}}$ [kg]	1017.32	5.5 %	1073.27	1149
$M_{\text{pressurant}}$ [kg]	1.51	20 %	1.81	-
$V_{\text{propellant tank}}$ [m^3]	1.169	1 %	1.181	1.175
$M_{\text{propellant tank}}$ [kg]	17.30	-	-	-
$M_{\text{pressurant tank}}$ [kg]	3.26	-	-	-
M_{PS} [kg]	32.58	10 %	35.83	-

Table 6: Estimated PS data with and without margins.

The propellant mass falls within reasonable margins compared to the actual value. The remaining difference is due to the fact that the total ΔV considered for the mission includes an additional 135 m/s reserve [43], which were not taken into account in the reverse sizing process. The similarity between the two masses of propellant can be interpreted as evidence that the reverse sizing process can be considered reliable.

Nevertheless, the total mass budget is still likely to be underestimated. With the value obtained, it would account for only 3.5% of the dry mass, which is significantly lower than in most cases. This discrepancy can be explained by the simplified tanks shapes but primarily by the reverse sizing methodology itself, which provides an approximate estimation. In particular, the transition between the blow-down and pressure-regulated systems was modeled in a simplified manner due to the lack of data, which may have led to deviations from reality. Moreover, titanium was assumed as the material for both tanks, but other materials could end up causing significant variations in the final result.

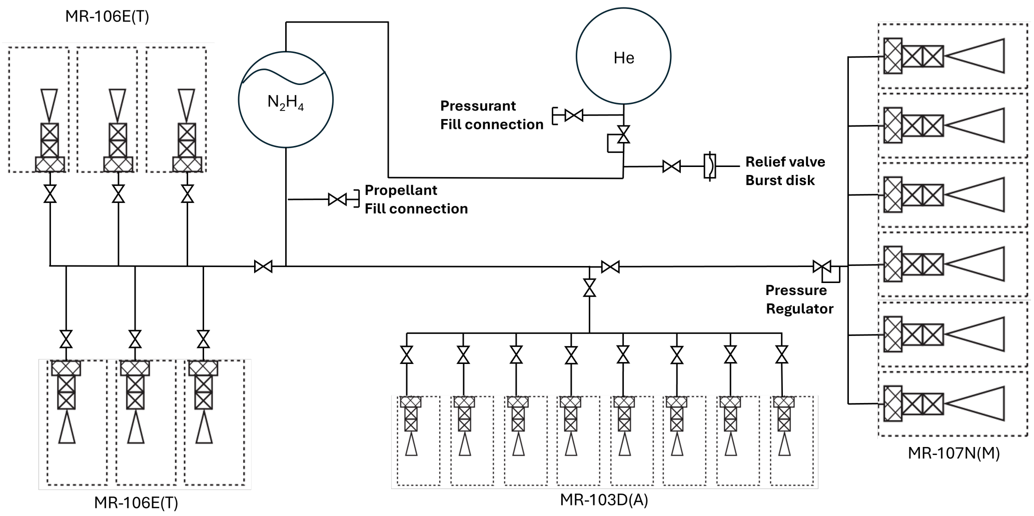
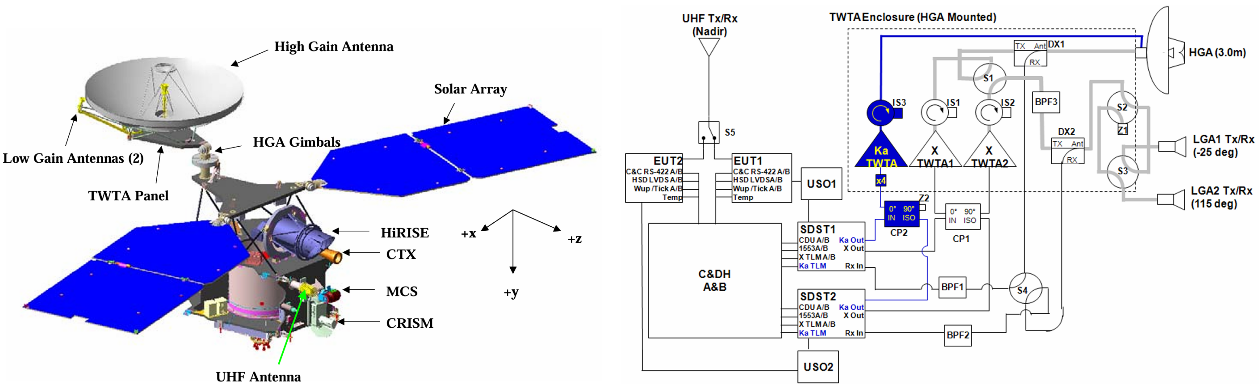


Figure 5: Propulsion architecture schematic.

3 Tracking, Telemetry & Telecommand Subsystem

The MRO mission has been designed to capture an unprecedented amount of high-resolution data and transmit it in the form of compressed raw data from the S/C to Earth. The full mission success criteria for data transfer was 26 terabits (Tb), the nominal PSP goal was 34 Tb, and over 50 Tb with the inclusion of Extended Mission (EM) and relay data. At the time of this report, MRO is in the EM and relay phase, having far surpassed both its expected mission lifetime and data transfer requirements (> 60 Tb).

Given the very high data transfer requirements and the Earth - S/C distance, an antenna with a very high gain is required for data transfer and communications. The S/C is also equipped with a pair of Low-Gain Antennas (LGAs) to enable low-data-rate communications to maintain basic Earth link during anomalies or critical phases. Additionally, the Electra system, operating in the UHF band, is included to achieve the MRO mission goal of acting as a primary telecommunication node within the Mars network. Figure 6 shows a schematic of the entire telecommunications architecture implemented on the MRO, along with the placement of the antennas on the spacecraft body.



(a) (b)

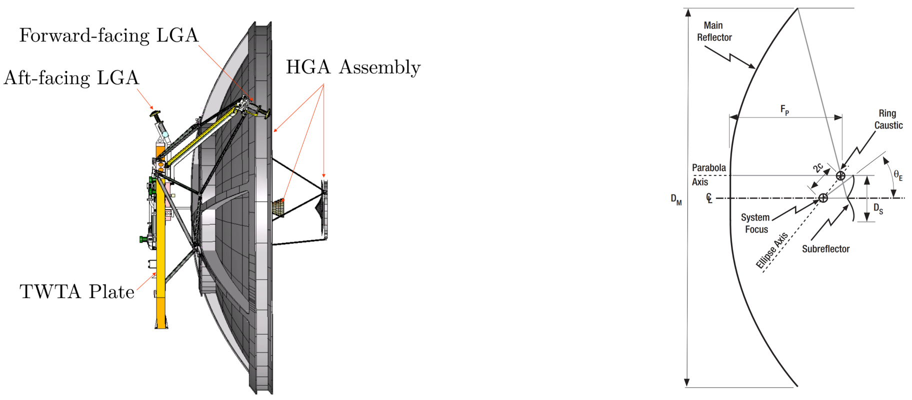
Figure 6: **(a)**: Configuration of MRO. **(b)** Telecommunication scheme of MRO [96].

3.1 Architecture and rationale

3.1.1 High-Gain Antenna System

The HGA serves as the primary means of communication between Earth and the S/C, addressing the high gain requirement. It is equipped with three traveling-wave tube amplifiers (TWTAs): two in the X-band (for redundancy), and one in the Ka-band (see section 3.1.4). The high gain required can be achieved by the use of reflectors or Direct Radiating Arrays (DRAs). While the latter provides the capability of electronic beam steering, it also introduces added complexity to the S/C, in addition to not being a technologically mature solution at the time of design. For similar reasons, along with considerations of reliability and signal integrity, a fixed parabolic reflector (shown in Figure 7a) was chosen over the alternative solutions such as deployable mesh antennas, which would have been susceptible to the stressing aerobraking conditions. The HGA is highly directional, and must point at Earth for data transfer. As such, a two-axis gimbal mechanism was implemented to ensure continuous communication.

The HGA has a paraboloidal main reflector with axial displaced elliptical (ADE) geometry (3 m in diameter) and an ellipsoidal subreflector (0.45 m in diameter), shown in Figure 7b. The feed, placed in the focus of the system, has two components: a corrugated horn X-band feed and a disc-on-rod Ka-band feed [40]. The choice of corrugated horn for the X-band feed is critical to lower the cross-polarization, including those arising from the reflector configuration [7].



(a) (b)

Figure 7: **(a)** The HGA, LGA, and TWTA [61]. **(b)** The ADE antenna geometry [44].

During aerobraking maneuvers, the HGA is locked in position in order to achieve the required drag surface, and telecommunications are happening through LGA1 [40].

3.1.2 Low Gain Antenna System

MRO is equipped with two Low-Gain Antennas (LGAs), designed for low-data-rate communication during critical mission phases such as launch, MOI, and safe mode operations. LGAs emit a broader radio beam, allowing for omnidirectional coverage at the expense of data rate. These horn-type antennas are essentially open waveguides with RF choke rings for pattern uniformity and side-lobe control. They are both mounted on the HGA, facing opposite directions. This configuration ensures continuous communication with Earth via the Deep Space Network (DSN), regardless of the spacecraft's orientation. The LGAs operate solely in the X-band frequency and do not support Ka-band capabilities. The forward-facing LGA1 is canted 25 degrees from the HGA boresight, whilst the aft-facing LGA2 is canted -115 degrees.

3.1.3 Electra

The Electra payload is a critical component for the relay operations of the S/C, and consists of a UHF transceiver and a low-gain UHF quadrifilar helix antenna. As for the NASA Mars Exploration Program (MEP), every future science orbiter will be required to host the standardized P/L, in order to provide relay and navigation services to both present as future spacecraft part of MEP. MRO was the first orbiter to incorporate the Electra payload [47]. The transmitting and receiving frequencies depend on which mode the P/L is running, with Full Duplex transmitting in the 435-450 MHz band, whilst receiving in the 390-405 MHz band. If the P/L is in Half Duplex, the P/L both transmits and receives between 390 and 450 MHz.

In comparison to the science P/Ls of MRO, Electra is more specifically designed to avoid electromagnetic interference (EMI), which, if generated, would cause a degradation of the communications for the DSN. Specifically, at the default relay return link frequency of 401.59 MHz, the P/L is designed to withstand the effects of EMI. Electra includes a single nadir-pointing (vertically towards Mars) UHF LGA. The antenna shares the nadir deck with science payloads, some of which are sensitive to UHF frequencies. This proximity can lead to electromagnetic coupling between the payloads and the antenna, distorting its gain pattern.

3.1.4 Bands

X-band: Most of the communication with the Deep Space Network (DSN) has been executed at the X-band, 8.439 GHz for downlink (DL) and 7.183 GHz for uplink (UL) transmission. The X-band was chosen due to its ability to reliable communicate at long distances, resist atmospheric interference, support high data transfer rates and its long-standing heritage in deep space communications [40].

Ka-band: MRO was equipped with a Ka-band (32 GHz) transmitter to evaluate its potential for high-rate data return thanks to its wider allocated bandwidth compared to X-Band. During the cruise phase, the Ka-band system showed promising performance; on October 31, 2005, up to 133 Gbits were received at record speeds of 6 Mbps [40]. Although the Ka-band is more susceptible to atmospheric losses due to its higher frequency, it is less affected by solar interference compared to the X-band, as confirmed by previous in-flight tests. Despite providing these advantages, a Ka-band exciter anomaly occurred on May 26, 2006, just before the start of orbital operations. The project chose not to risk activating the system, and the operational demonstration was canceled for both the primary and extended missions [40].

UHF: UHF bands (435-450 MHz Full Duplex mode, 390-405 MHz only receiving mode) were selected for short-range relay communications between MRO and orbiters, landers, and rovers due to its high performance in the thin Martian atmosphere. At the aforementioned frequencies, atmospheric losses on Mars are almost negligible, even during high atmospheric activity periods. Additionally, the use of UHF ensures full compatibility with the Proximity-1 protocol, NASA's standard for relay operations at Mars [15].

3.1.5 Ground Section

The DSN was selected as the main communication network for the MRO mission, since it was specifically designed to support interplanetary missions by adopting high-gain antennas and providing the required link power to compensate for the stringent free-space losses caused by the large Earth–Mars distance. The DSN is composed of three stations at widely separated longitudes to provide continuous view from deep space (Goldstone in the USA, Madrid in Spain, and Canberra in Australia) [40].
These stations employ different types of antennas. The 70 m diameter antennas are primarily used during critical phases like Mars Orbit Insertion or safe mode operations due to their higher gain, pointing precision and sensitivity (signal acquisition is possible even at extremely low received power levels from the LGAs), as quantitatively described in the technical report [98].
the 34 m beam-waveguide (BWG) antenna subnet, and the 34 m high-efficiency (HEF) antenna subnet are used for routine communication sessions, such as cruise, science and relay phases due to the greater flexibility, lower operational cost, and easier scheduling provided by said subnets [40, 91].

Other communication networks were used during the LEOP phase, such as the Uchinoura Space Center (Japan), which received the first signal after separation [40].

3.1.6 Encoding and Modulation

The modulation of a carrier signal is performed inside the Small deep space transponder (SDST), employing Phase-shift keying (PSK) instead of Frequency-shift keying (FSK), as PSK represents bits by shifting the phase of a single carrier frequency, whereas FSK uses multiple frequencies to represent bits (wider bandwidth), which is less suited for a deep space link. Binary phase-shift keying (BPSK) and Quadrature phase-shift keying (QPSK) are used at different phases and for different operational requirements [40]:

- **BPSK**: in a sub-carrier, or directly in the X-band signal carrier, the phase of the signal is shifted between 0° or 180°, to represent the bit state 0 or 1, BPSK is used in low-data-rate, emergency, or recovery phases.

– **QPSK**: four phase-shifting are used (0°, 90°, 180°, or 270°) to represent four possible bit pairs (00 01 10 11), QPSK is used for high-data-rate transmission during regular science operations[40].

The distance between Mars and Earth and the high data rate impose a very low signal-to-noise ratio (SNR), therefore powerful Error-correction code (ECC) are needed. X-band and Ka-band communication systems employed the following coding schemes[40]:

– **Turbo Codes**: support bit rates above 32 kbps with block lengths of 8920 bits and code rates of 1/2, 1/3, 1/6. They are favored in telemetry and science data links on X and Ka bands due to their near Shannon-limit performance, allowing high protection to high data rate for a given bandwidth. Turbo code encoding is used for robust error correction over long distances [96].

– **Reed-Solomon (RS)**: Up to 6.6 Mbps, used primarily for very high data rates in situations where MRO is close enough to Earth that coding gain is less important, due to an already strong signal, such as early cruise and Mars-Earth closest approach (low SNR);

– **Convolutional Concatenated**: Combination of (7, 1/2) convolutional and RS codes, used in interplanetary mission where power is limited. This code is employed in emergency mode (34.38 bps) in short frames, and up to 3.3 Mbps in long frames during Ka-band downlink;

3.2 Con-Ops and Phases correlation

During the LEOP phase, only the LGAs were used to maintain basic communication, once the HGA was deployed, a stronger link with ground was established and the cruise phase began. During cruise, all communication systems were tested, but only LGA and HGA were used for routine data exchange, while most Ka-band experiments took place. In critical maneuvers such as TCMs and MOI, only the LGAs were used due to HGA pointing constrains. During aerobraking, communication alternated between the two systems: the LGA near pericenter (highest criticality), and the HGA along the rest of the orbit. The Ka-band communications demonstration was planned to conduct activities during solar conjunction to compare simultaneous X and Ka-band telemetry downlinks. The DSN supported one 8 hour pass per day to a 34-m antenna during this period. Once in its operating orbit, MRO went through periodic occultations behind Mars, preventing Earth contact for about one third of orbit's time. Nevertheless, during the orbital mission, data was transmitted for 10–11 hours daily. In the PSP, MRO was allocated two 34-m passes per day (8 hours each) and three 70-m passes per week. The Electra UHF relay system became operational only in the dedicated relay phase [40].

System	Phases and ConOps	Band	Rationale
HGA	Cruise, Aerobraking, Science and Relay phases, Solar Conjuction (Ka-band tests)	X-band / Ka-band	High Precision High Datarate Dual-band (X/Ka) Capability
LGA	Safe mode, Launch Phase, TCMs , Aerobraking, MOI	X-band	Low Power Omnidirectional Coverage Emergency Link Support
Electra	Early Cruise Test, Relay Phase	UHF	Wide Beamwidth Mars Network Relay Node

Table 7: Relation with Operations, phases, modes

3.3 Reverse Sizing

This section aims at describing the reverse sizing procedure of the TMTC subsystems, focusing exclusively on the X-band communications with Earth during PSP, the phase with the highest data rate, under the worst-case operation conditions. This section includes the input data and the assumptions for the analysis, followed by a presentation of the results for both uplink and downlink.

3.3.1 Downlink Sizing Data and Assumptions

The X-Band frequency used to send data from MRO to Earth is f_{dl}=8439.55 MHz according to guidelines provided by ITU, with a data rate R_{dl} = 500 kbps [40]. Taking into account the most critical condition, Earth-Mars distance was assumed at its highest (401.4 million km) during near-solar conjunction. The available power of the transmitting 3-m wide HGA is P_{tx}=100 W [40], with an electrical power to be provided to the antenna of 140 W, considering the amplifier efficiency recovered graphically (η_{amp}=0.6) and an aperture efficiency of η_{HGA}=0.65 due to its geometry. The system uses a BPSK modulation scheme (α_{mod}=1) and a turbo code (8920, 1/2) encoding (α_{enc}=2). While 1/6 rate turbo code would be more robust, the ground decoding only supports data rates up to 1.5 Mbps using turbo code [96]. The 10^{-5} BER leads to a minimum energy bit over noise power spectral density of 0.5 dB [106]. Some assumptions have also been taken to compute the losses affecting the link budget. The cable losses were estimated with a typical value of −1.0 dB (reported as minimal [40]) and the total pointing losses were taken from literature equal to −0.1 dB [99]. The atmospheric attenuation has been taken into consideration based on values available in the literature for an elevation angle of 10° and average clear skies (CD = 0.25) [23]. The Sun-Earth-Probe (SEP) angle was assumed sufficiently large ($> 1.5°$) such that solar noise contribution is negligible. The system noise temperature was calculated using the procedure outlined in [91].

3.3.2 Uplink Sizing Data and Assumptions

The UL sizing procedure follows the same steps as the DL, with changes to some assumptions and data. The same distance is assumed as DL, as the same worst case cited above is considered for both UL and DL. The UL frequency is f_{ul} = 7.183 GHz, with a data rate of R_{ul} = 4 kbps, taking into account a margin around values found in the literature [40]. The ground station DSS 25 in Goldstone, which has an aperture of 34-m, an aperture efficiency of $\eta_{\text{tx}} = 0.65$, and a power capacity of 20 kW [99]. It was assumed that a power of P_{tx} = 2 kW is reserved for communicating with MRO. The modulation scheme for the UL is BPSK ($\alpha_{\text{mod}} = 1$), and 1/2 rate convolutional encoding ($\alpha_{\text{enc}} = 2$) is employed. For UL data transmission, a minimum bit error rate (BER) of 10^{-7} is set. Cable losses are assumed $L_{\text{c}} = -2$ dB. The atmospheric loss is estimated using $A_{\text{zenith}} = -0.04$ dB from[23] and an assumed elevation of 10°. The noise temperature was assumed as T_{rx}=290 K following the guidelines provided for the subsystem sizing.

3.3.3 Results Analysis

The results for both UL and DL are reported in Table 8. The overall link budget shows that the subsystem is well-sized for a reliable link both in UL and DL. As expected, free space losses are the driving losses as one of the main factors in MRO's link budget, due to the large distance S/C - Earth. The computed gains of the HGA antenna for the X-band for both UL and DL are similar to the ones reported in the literature [40].

In UL, high power is needed in order to compensate for the low gain of the receiving antenna on the S/C. The resulting E_b/N_0 and SNR are both well above the required minimums, granting solid reliability.

The DL case is much more critical, due to less available power. In addition, the data rate is considerably higher. These aspects make communications with Earth much more challenging. The computed values of E_b/N_0 and SNR are expectedly lower than the UL case. Both values are still above their respective minimums, but by a smaller margin than in the previous case, which makes the

DL sizing less reliable than that of UL, as any misapproximation could lead to overstepping the link budget constraints. The DL sizing would need more detailed modeling and data in order to obtain a more reliable result. In order to enhance the results computed in the DL case, one could reduce the data rate, increase the available power, or increasing gain of either antenna.

	UL	DL	Unit
Transmitted power (P_{tx})	2000	100	W
TX antenna gain (G_{tx})	66.28	46.60	dB
RX antenna gain (G_{rx})	45.20	67.68	dB
Cable losses (L_{c})	-2	-1	dB
Pointing losses (L_{point})	-0.21	-0.1	dB
Atmospheric losses (L_{atm})	-0.22	-0.22	dB
Free space losses (L_{FS})	-281.02	-282.42	dB
Bandwidth (B)	40.49	61.46	dB
Net datarate (R_{net})	4	500	kbps
Gross datarate (R_{gross})	8	1000	kbps
RX system noise temperature (T)	290	33	K
Minimum Eb/N0 (with 3 dB margin)	8	3.5	dB
Minimum SNR (with 3 dB margin)	6.54	2.04	dB
Eb/N0	26.16	6.31	dB
SNR	24.70	4.85	dB

Table 8: MRO link budget.

3.4 Mass and Power Budget

The TMTC subsystem mass and power budget are summarized in the table below considering the different components involved. The reported values highlight the importance of this subsystem into the design process of the whole satellite. The choice of gimbals for the HGA is justified by the need of continuous communications, even if it strongly affects the total mass budget. The power required by TWTAs is then necessary to cover the long distance with respect to Earth.

Assembly	**Mass [kg]**	**Power [W]**
X-band transponder	6.4	16
TWTAs	12.1	253
X-band & Ka-band antennas	22.6	-
HGA gimbals	45.0	14
Waveguides	8.3	-
Ultra-Stable Oscillator	1.7	5
UHF subsystem	11.5	71
Telecom total	**107.7**	**359**

Table 9: TTMTC subsystem mass and power breakdown [40].

Change log	
Issue	*Comments*
§4.3	pp. 24: Table 11 revised
§4.4.2	pp. 25: unit of measurement corrected
§4.4.4	pp. 26: unit of measurement corrected

4 Attitude and Orbit Control Subsystem

The Attitude and Orbit Control Subsystem (AOCS) provides the hardware and the software capabilities to perform attitude measurements, estimation, guidance and control required across all of the control modes of the mission. The AOCS allows to perform corrective maneuvers on the trajectory to Mars, while keeping the solar arrays and the HGA pointed toward the Sun and Earth respectively to satisfy the science requirements. This subsystem relies on two star trackers, sixteen Sun sensors, and two inertial measurement units to determine the MRO attitude. A set of four reaction wheels (RW) and eight Reaction Control System (RCS) thrusters are used to control the attitude [72, 73].

4.1 Pointing Performance Related Requirements

The design of AOCS was driven by two main performance requirements: target relative pointing and pointing stability. The first defines how well a body-fixed vector must point at a specified target on the surface of Mars, ensuring that it is captured in the instrument FOV. The second specifies how still the spacecraft must remain over a specified period of time, ensuring that the image quality meets the science objective [51]. The stability requirements for each imaging instrument are summarized in Table 10, along with the time interval in which this accuracy shall be maintained. It can be seen that the HiRISE requirements are, by far, the most challenging with respect to the other instruments. This is also valid for target pointing requirements, for which the HiRISE boresight shall point to a target on Mars within 0.7 mrad (3-sigma) in the FOV cross-track direction and 2 mrad in down-track direction.

Instrument	**Magnitude**	**Window**
HiRISE	0.0032 mrad	12 msec
CTX	0.020 mrad	2.5 msec
CRISM	0.032 mrad	200 msec
ONC	0.033 mrad	300 msec
MCS	3 mrad	16 sec

Table 10: MRO science instrument pointing stability requirements [51].

A crucial aspect to consider in the design of the AOCS is the simultaneous operation of multiple instruments, which makes it necessary to account for and correct (as much as possible) the instrument misalignment [51]. The pointing requirements of the Tracking, Telemetry and Telecommand Subsystem (TTMTC) and the power subsystem are mainly handled by the gimbal mechanisms of the HGA and the solar arrays respectively, and thus are not drivers for the AOCS design.

4.2 Architecture and Rationale

Even if the most challenging requirements derive from HiRISE, the architecture design must take into account all the instruments used for science objectives, as well as the conditions for proper functioning of the spacecraft. MRO is designed to support continuous nadir-oriented operations at

Mars and periodic maneuvering to obtain high resolution images of targets. The orbiter is able to roll and capture data ±30° cross-track of nadir, which is operationally held below 10° as much as possible to meet scientific requirements. MRO is equipped with strongly reliable sensors and actuators, always granting redundancy in case of failure. In the following paragraphs the different components of AOCS are specifically described.

4.2.1 Sensors

MRO is equipped with two Honeywell Miniature Inertial Measurement Units (MIMUs), two Galileo Avionica A-STR star trackers and sixteen Sun sensors.

MIMU: each device is a combination of three accelerometers and three ring laser gyroscopes (one gyro and one accelerometer for each axis). The accelerometers measure the accelerations to provide information about the firings of the thrusters, while the gyroscopes measure the rotational velocity of the spacecraft. MIMUs allow also to estimate the orbiter orientation for brief periods when the angular rate is too high for star trackers [73]. These devices are highly reliable, ensuring less than 2200 Failure in Time (FIT), meaning that it can have a maximum of 2200 failures over a period of one billion hours [35]. The second MIMU is equipped for backup purposes in case of failure of the primary device.
Sun Sensors: they are distributed on the solar arrays and orbiter bus, allowing the spacecraft to locate the Sun. Only eight of the sixteen sensors are used, while the others are equipped for redundancy. They can only provide information about the presence of the Sun in their FOV. Combining the information of all the sensors, the on-board computer can roughly estimate the position of the Sun. These devices are typically used during the waking-up of the spacecraft after launch or safe mode, giving the spacecraft enough information to re-orient towards and get power from the Sun. However, information from Sun sensors alone is not sufficient to estimate precise position and attitude [73].
Star Trackers: the star trackers provide full knowledge of the orbiter orientation, identifying the stars around and matching them with its database. Due to the crucial role of this device, a second redundant star tracker is included, in case of failure. This device continuously takes digital images of the surrounding space and compares them with an on-board catalog until it can identify the stars in the images [51]. The autonomous A-STR star tracker is strongly reliable (TRL-9) and offers high performances with its CCD detector. It is a medium FOV tracker (16.4° x 16.4°), able to identify up to 10 stars with a tracking rate of 2°/s. Despite its reduced dimensions (195 x 175 x 290.5 mm), this device is affected by very low random errors, i.e. Noise Equivalent Error (NEA), of less than 25 arcsec for pitch and yaw, and less than 230 arcsec for roll [52].

4.2.2 Actuators

The actuators allow to translate attitude commands into actual control motion of the orbiter. The MRO is equipped with four Honeywell Constellation Series HR16 RWs and eight small Aerojet MR-103D thrusters, comprising the RCS.

RW: this assembly is composed of three wheels mounted perpendicularly to each other (one for each rotational axis, commonly aligned along the body-frame axis), with a fourth wheel mounted in a skewed direction as a redundant unit in the event of failure of one of the primary wheels. RWs are used during all the mission phases [43]. They provide slow and steady turns, which are suitable for high-resolution images required for science objectives. Each wheel has a mass of about 12 kg and a maximum angular velocity of 6000 rpm. The torque produced can be regulated controlling the wheels with DC brushless electric motors, ensuring low jitter disturbance and reaching a maximum reaction torque of 0.4 Nm. The power consumption of each unit requires at their peak 195 W when applying

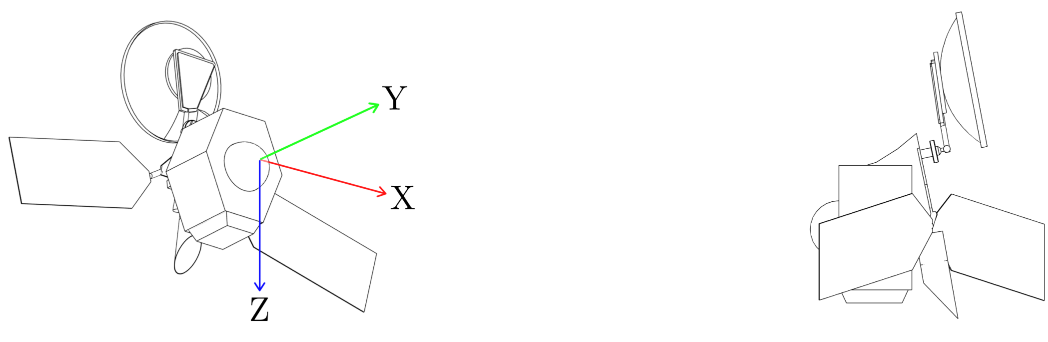
torque, and about 22 W in their steady state at 6000 rpm.
RCS: this system relies on eight thrusters, coupled in pairs, that together can spin the spacecraft with minimal residual ΔV. They are suitable for quick turns and get new orientations in few seconds. Redundancy is granted, in fact the RCS is still able to work in case of failure of one thruster in any paired couple, even if some lateral velocity is given [73]. The RCS is also used for the desaturation procedure of the RWs, reducing their accumulated angular momentum. In fact, since MRO is only symmetric along one axis, a non-zero net torque is applied to the orbiter due to various disturbances nearly at all times, which is counteracted by RWs until they are saturated. In general, the RCS is employed whenever the RWs control system is not suitable or unavailable, such as safing events, TCMs, MOI and aerobraking [43]. Each thruster has a mass of 0.33 kg, a nominal thrust of 0.9 N. The specific impulse is estimated between 209 and 224 seconds, while the minimum impulse bit stays around 0.27 Ns [2]. While the placement of thrusters are not publicly available, they are presumed paired and equidistant from the shifting axis of the Center of Mass (CoM).

4.2.3 Flight Control Software

The attitude estimation is performed by a 6-state extended Kalman filter. The on-board computer combines the quaternion estimates from the star trackers with angular speed measurements from MIMU to produce a 3-state error estimate and a 3-state gyro bias estimate. The uplink system provides ephemerides and pointing commands to the spacecraft, which processes and passes them on to the other components of the AOCS as attitude commands [51].

4.3 Pointing Budget and Modes Correlation

The spacecraft body-fixed frame is defined such that +Z is aligned with the normal direction of the nadir deck, +Y is oriented along the centerline of the propellant tank and the MOI thrusters, and +X is defined by the cross product of +Y and +Z, as shown in Figure 8a. The origin of the frame is located at the geometric center of the launch separation adapter plate. Both the HGA and solar arrays are equipped with gimbals able to rotate along their inner and outer rotation axes. Different attitude configurations are adopted throughout the mission, depending on the specific requirements of each phase and operational mode. The following paragraphs describe the main attitude configurations during different modes of the main mission phases.



(a) (b)

Figure 8: **(a)** Simplified model of MRO. **(b)** MRO as seen from the drag direction in PSP.

Checkout Mode (CM): right after the separation from the launcher, the orbiter was released with an angular velocity of 0.3°/s around X axis. After a series of appendage-deployment activities, MRO was settled in a preset inertial-fixed acquisition attitude. In this configuration the -Y axis is pointed to the Sun with a bias error of 15° around the X axis, which is obtained by the cross product of Sun and velocity direction vectors. The solar arrays gimbal angles were null, while the inner and outer rotation angles of the HGA gimbal are respectively set to 180° and -45° [111].

Sun-point Cruise Mode (SCM): about three days after launch the spacecraft transitioned into this configuration, pointing the -Y axis to the Sun. The -X axis is obtained by the cross product between the Sun and Earth direction vectors. The solar gimbal angles are null, the inner angle of the of the HGA is fixed at 180°, while the outer angle is the result of the Sun-probe-Earth angle minus 90°. This configuration ensures that the Earth is always in the YZ plane and allows the HGA to easily track the Earth maintaining its gimbal angle constant [111].

Spread-Eagle Mode (SEM): two months before MOI, the orbiter assumed this attitude, with -Y axis pointed toward Earth and +Z axis result of the cross product between Earth and Sun direction vectors. Both solar arrays gimbal angles are set to zero, while HGA gimbal inner and outer angles are respectively set to 180° and -90°. This configuration allows to maximize the cross area of the HGA and exploit air drag during MOI. This attitude is also used for TCMs and AMD events [111], exploiting MRO's asymmetry to aid in attitude control.

Aerobraking Mode (AM): this configuration is analogous to the previously described spread-eagle mode, except with lower pointing accuracy due to harsh conditions.

Global Mapping Mode (GMM): this is one of the three observation modes assumed during the PSP. The instruments involved, MCS and MARCI, require nadir pointing, low data rate and continuous operations. This means that the +Z direction is aligned with the nadir one, while the HGA and solar arrays gimbal angles vary to track respectively the Earth and the Sun.

Survey Mode (SUR): this configuration is assumed to allow CRISM, CTX and SHARAD to work properly. The attitude is similar to GMM, but requires much more precise pointing accuracy.

Target Investigations Mode (TIM): this observation mode includes the usage of the CTX, the CRISM and HiRISE. The target investigations often requires off-nadir rolls up to 30° for HGA and solar arrays constraints. However, rolls over 9° are limited to one per day due to the effects on MCS studies. The nominal attitude is still the same as the other observation modes (+Z axis aligned with nadir direction), but the HiRISE requires incredibly high pointing accuracies as reported in Table 10.

Safe Mode (SAFE): if an anomaly occurs, MRO switches to Safe Mode, a stable configuration. In this mode, LGA1 is pointed toward Earth, and the solar arrays are Sun-pointed, with the spacecraft –Y axis tracking the Sun. Sun and Earth ephemerides loaded before launch are used to determine the Sun–probe–Earth angle and guide the initial pointing of the LGA. Star trackers can be used for Sun acquisition. If star trackers are unavailable, and the only attitude knowledge comes from Sun sensors, the spacecraft rotates about the –Y axis, causing the LGA boresight to trace a cone relative to Earth, thus signaling the presence of an anomaly to the ground stations [103].

Mode	**Main Tasks**	**APE [°]**
Checkout	Detumbling, appendage-deployment, initial attiude determination	< 15
Sun-Point Cruise	HGA communications, solar arrays Sun-pointing, calibrations	< 0.12
Spread-Eagle	MOI preparation, wheels desaturation, air drag and asymmetry exploitation	< 5
Aerobraking	Air drag exploitation	< 7
Global Mapping	MARCI and MCS continuous operations	< 0.1
Survey	CRISM, CTX and SHARAD extended nadir observations	< 0.1
Target Obs	CTX and CRISM operations, HiRISE high precision target observations, roll maneuvers	< 0.01
Safe Mode	LGA communications, solar arrays Sun-pointing	< 10

Table 11: Control modes and pointing budgets

4.4 Reverse Sizing of the AOCS

In this section, the attitude control system of the MRO is analyzed starting from mission requirements and performance objectives, considering how sensors, actuators, and environmental disturbances interact. Through a reverse sizing approach, each component is iteratively studied to ensure it can reasonably meet operational needs across all mission phases. This process justifies design choices in terms of sizing, redundancy, and allocation of key resources such as mass, power, and data.

4.4.1 Mass and Inertia Properties

Using the simplified model, with remaining propellant mass corresponding to that of the PSP initialization, the following physical properties are obtained. While they are varying in reality, these values are held constant across calculations for different phases.

Parameter	Value	Unit
Mass	1897	kg
$I_x = I_{\max}$	2898	$\mathrm{kg\,m^2}$
$I_y = I_{\min}$	1961	$\mathrm{kg\,m^2}$

Table 12: Physical properties relevant to AOCS sizing.

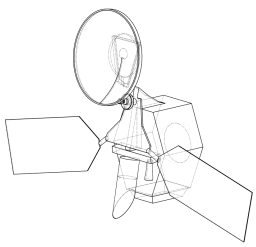


Figure 9: The simplified MRO model.

4.4.2 External disturbances

During the mission, the interaction with external disturbances affected the attitude dynamics of the spacecraft. The main ones are the following: gravity gradient, atmospheric drag and solar radiation pressure (SRP). The torques, resulting from these disturbances, are evaluated with the following equations:

Disturbance	Reference formula	Result [N m]
Gravity gradient	$T_{\mathrm{GG}} = \frac{3\mu_{\mathrm{Mars}}}{2R^3_{\mathrm{orb}}}(I_{\max} - I_{\min})\sin(2\theta)$	1.445×10^{-3}
Solar radiation pressure	$T_{\mathrm{SRP}} = \frac{F_S}{c} A_{\mathrm{Sun}}(1+q)\cos(\mathrm{i})\ \lVert\vec{r}_{\text{GC-CM}} \times \hat{n}_{\mathrm{SA}}\rVert$	9.08×10^{-4}
Aerodynamic drag	$T_{\mathrm{d}} = \frac{1}{2}\rho C_d A V^2(c_p - c_g)$	5.7×10^{-3}

Table 13: Disturbance torques using MRO and Mars environment parameters.

For all the calculations, the worst case scenario is assumed, so the angle θ in the gravity gradient formula is considered the maximum off-nadir pointing angle of the HiRISE (30°). The incident angle in the expression of the SRP is assumed 0°, while the relative velocity V, used for the drag torque is assumed equal to the average orbital velocity of the spacecraft, 3405 m/s. For the majority of PSP, it is reported that the +X side of MRO is exposed to the drag [111]. The worst case, i.e. the maximum exposed area, is driven by the keep-out-zones of the gimbals [70], replicated using the simplified model as shown in Figure 8b.

4.4.3 Reaction Wheels

The four RWs are disposed, inside the spacecraft, in a configuration with three rotors aligned with three orthogonal axis and the fourth with equal components along these axis. This configuration can be expressed in matrix form with each row having a unitary versor in its corresponding axis, and $1/\sqrt{3}$ throughout the fourth column, representing the fourth wheel.

Attitude station keeping: starting from the total torque applied on the spacecraft, the torque of each single reaction wheel, to counteract external disturbances and keep the MRO aligned, is evaluated considering the worst case, with external torque concentrated along one of the three orthogonal axis: $T_{\mathrm{RW\_disturbance}} = \frac{T_d}{1+\frac{1}{\sqrt{3}}} = 4.6 \times 10^{-3}\mathrm{N\,m}$ (for PSP case). Under the influence of external disturbances, the angular momentum accumulated by a single reaction wheel results: $h_{\mathrm{RW}} = T_{\mathrm{D}} \cdot t_{\mathrm{orb}}$, so the period of time after which it is necessary a desaturation maneuver can be estimated, and is evaluated in the next section.

Slew maneuver: during the mission the ADCS subsystem must be able to perform slew maneuvers and realign the pointing direction with the target one. The slew maneuver with RWs, is performed with a constant acceleration and breaking. Considering the maximum torque, the slew rate go above the maximum value imposed ($\dot{\theta}_{\mathrm{slew,max}} = 0.5°/\mathrm{s}$), so a new torque for the maneuver is computed, $T_{slew} = 0.16\mathrm{N\,m}$, lower than the maximum, so the time required for the slew maneuver:

$$t_{\mathrm{slew,RW}} = \sqrt{\frac{4 \cdot \theta_{\mathrm{slew,max}} \cdot I_{\max}}{T_{\mathrm{slew}}}} = 466.10s$$

The maximum angular momentum, reached during the slew maneuver, results:
$h_{\mathrm{slew,max}} = T_{\max} \cdot t_{\mathrm{slew,min}} = 25.29\mathrm{N\,m\,s}$, lower than the maximum angular momentum that can be stored by these RWs. Then the mechanical power provided by the RW is calculated and converted in the electrical power requested using their electric-to-mechanical efficiency [36].

4.4.4 Thrusters

The configuration of thrusters is characterized by four banks, each with two thrusters. The reverse sizing is performed considering that only four thrusters contribute to the total torque and they are disposed symmetrically with respect to the center of mass. The sizing of thrusters is realized considering three main attitude control scenarios:

Attitude station keeping: the necessary torque to counteract the effects of external disturbances and maintain the desired pointing, results from the following equation: $F_{\text{th}} = \frac{T_D}{nL} = 9.08 \times 10^{-4}$ N (for PSP case).

Slew maneuver: with thrusters to get a desired slew angle, can be achieved in different maneuvering sequences: acceleration, coasting and breaking. The maximum thrust F_{th} is equal to 0.9 N, taken from datasheet, the resulting torque applied is: $T = nF_{\text{th}}L$, the time of the maneuver is given by:

$$t_{\text{acc}} = t_{\text{brake}} = \frac{\dot{\theta}_{\text{acc,max}} I_{\text{max}}}{nFL} \qquad t_{\text{coast}} = \left(\theta_{\text{slew,max}} - \frac{nF_{\text{th}}L}{I_{\text{max}}} t_{\text{acc}}^2\right) \cdot \frac{1}{\dot{\theta}_{\text{coast}}}$$

with $t_{\text{slew}} = t_{\text{acc}} + t_{\text{coast}} + t_{\text{brake}} = 363.51\text{s}$. The propellant mass for a single slew maneuver can be evaluated with the specific impulse I_{sp} and the total impulse I_{tot}: $M_{\text{prop}} = \frac{I_{\text{tot}}}{I_{\text{sp}}} = 0.0031\text{kg}$, with $I_{\text{tot}} = t_{\text{slew}} \cdot F_{\text{th}}$.

Desaturation: considering data of thrusters and RWs, from the previous section, the time required to desaturate using the maximum thrust is higher then typical value reported in literature, so the thrust to desaturate is computed as follow: $F_{\text{desat}} = \frac{h_{\text{max}}}{nLt_{\text{desat}}} = 0.4167\text{N}$. The number of desaturation events can be evaluated computing the period before reaching saturation and the duration of the phase of interest, the propellant mass required can be estimated using the aforementioned formula but with a total impulse given by: $I_{\text{tot}} = n \cdot t_{\text{desat}} F_{\text{desat}}$.

4.4.5 Mode Correlations

In the following tables, the correlation between the operational modes and the most relevant AOCS parameters and technologies is presented. Only the dominant disturbance sources are listed, and for sensors and actuators, only the primary devices involved in each mode are highlighted.

Phase	Mode	Dist.	Act.	Sens.	Slew Sizing	Desaturation Sizing
LEOP	CM	N/A	RW	Sun S., MIMU	N/A (Detumbling)	N/A
Cruise	SCM	SRP	RW	Sun S., MIMU	$n_{\text{slew,day}} = 6$ $E_{\text{day}} = 46.41\text{W h}$	$n_{\text{des,day}} < 1$ $M_{\text{prop,tot}} = 0.1\text{kg}$
	SEM	SRP	RW, RCS	MIMU, Star T.		
ATP	AM	Drag, SRP	RW, RCS	MIMU, Star T.	$n_{\text{slew,day}} = 29$ $E_{\text{day}} = 258.41\text{W h}$	$n_{\text{des,day}} = 5$ $M_{\text{prop,tot}} = 7.08\text{kg}$
PSP	GMM	Drag	RW	Star T., MIMU	$n_{\text{slew,day}} = 31$ $E_{\text{day}} = 276.96\text{W h}$	$n_{\text{des,day}} = 6$ $M_{\text{prop,tot}} = 26.42\text{kg}$
	SUR	SRP	RW			
	TIM	Grav.	RW, RCS			

Table 14: AOCS parameters by phase and mode.

In Table 14, the slew maneuvers are considered to be performed exclusively using RW, while desaturation is achieved solely by using the RCS thrusters. For completeness, a propellant consumption of 0.0031 kg is designated for a full slew maneuver relying only on thrusters. It should be noted that the $M_{\text{prop,tot}}$ listed in the table refers to the total propellant mass consumed during all of considered phases only to desaturate the S/C.

4.4.6 Mass, Power and Data Budgets

Component	**Mass** [kg]	**Power** [W]	**Data Rate** [kbps]
Reaction wheel assembly (RWA)	48 (12×4)	68.53	~ 0
RCS	32.74	-	~ 0
IMUs	9.2 (4.6×2)	25	23.04
Star trackers	7.2 (3.6×2)	8.9	2.56
Sun trackers	3.4 (0.215×16)	0 (Passive)	~ 0
Total	101	102	~26 (~31 with margin)

Table 15: Mass, power, and data budgets of the AOCS, including redundancy.

The data rates are calculated using values (during slew) found in similar spacecraft, and supporting data found in the literature, such as float precisions and refresh rates. It is assumed that sensor outputs are 3+3 state matrices and attitude quaternions. A margin of 20% is applied to the total data rate to account for housekeeping data. The power for the RWA, reported in the table above, is computed to realize a slew maneuver of 180° with a single RW.

5 Thermal Control Subsystem

The Thermal Control Subsystem (TCS) controls the heat exchange in MRO to maintain the temperature of each component within their acceptable temperature ranges. Given its 3pm Local Solar Time (LST) Sun-synchronous orbit [29], MRO experiences less thermal variation compared to orbits closer to local noon, however, the spacecraft still experiences high solar exposure throughout the mission's duration. During nominal operations, MRO also encounters eclipses that result in large temperature swings. Furthermore, throughout the aerobraking phase, MRO is subjected to very high mechanical and thermal loads. For these reasons, the TCS of MRO employs a wide suite of techniques for thermal control. Said techniques and the specifics of subsystems employing thermal control are outlined in the following sections.

5.1 Thermal Requirements of Onboard Systems

The S/C components are characterized by an operational and a survival temperature limit. The most critical values of each subsystem are reported in Table 16:

	Temperature [°C]				
Component	**Min. Surv.**	**Min. Oper.**	**Max. Oper.**	**Max. Surv.**	**Notes / References**
Instruments					
MARCI	-80	-40	70	100	[57]
HiRISE	—	-110	20	—	[9]
CRISM	—	-163	-148	175	[109, 79, 19]
CTX	—	17	25	—	[37]
SHARAD	—	-50	100	—	Estimated
ONC	—	-30	40	—	Estimated* [45, 55]
TTMTC					
Antennas	-120	-100	100	120	Typical [49]
Antenna gimbals	-50	-40	80	90	Typical [49]
AOCS					
RWs	—	-30	70	—	[36]
Sun sensors	—	-45	85	—	Estimated* [1]
Star trackers	—	-30	60	—	[52]
MIMUs	—	-30	65	—	[35]
Propulsion					
Propellant tank and lines	5	15	40	50	Hydrazine, Typical [49]
Power					
Solar arrays	—	-150	175	200	[85]
Batteries	-25	-10	18	25	[71, 104]

Table 16: Operational and survival temperature ranges for MRO components.
*: based on similar equipment. —: unavailable or inapplicable.

The propulsion system's survival and operational temperatures exceed those of the rest of the S/C, necessitating dedicated heaters. Some instruments also operate outside the S/C's standard range. CRISM needs extreme cold for infrared (IR) imaging; CTX functions within a narrow, and comparatively higher temperature range than the S/C as a whole; and HiRISE is susceptible to thermal damage at high temperatures. These conditions are managed using cryocoolers, heaters, and radiators.

5.2 Architecture and rationale

To meet the thermal requirements across components, the system incorporates a combination of active and passive elements based on conduction and radiation. Their roles are briefly described below.
Radiators: used for dissipating heat from the heat-generating components MRO into deep space. Radiators are, when possible, directly attached to said components, absorbing their excess heat via conduction and dissipating it via radiation. Among the types utilized on MRO are aluminum, silver teflon and painted radiators [102].
Surface Coatings: these passive elements are used to modify the effects of radiation on the S/C surfaces, allowing to selectively determine how much heat is absorbed or reflected from, or radiated to the environment [82].
Multi-Layer Insulation (MLI) or Thermal Blankets: typically made of multiple layers of Mylar or Kapton separated by spacers, their main objective is to create an insulating layer around the component or spacecraft. They also incorporate surface coatings depending on thermal requirements.
Heaters: they are formed by wires capable of releasing heat due to their high electrical resistance. They are directly placed on and around units of MRO that require heating during the mission [82].

5.2.1 Payload

The instruments on board require thermal control components to work properly and maintain the operating temperatures listed in Table 16. CRISM operates at very cold temperatures to limit background radiation, being an IR instrument. While passive cooling is involved in keeping the housing of CRISM thermally stable via a graphite epoxy radiator [64], the solar-exposure-rich orbit of MRO necessitates some active cooling. This was achieved using three redundant cryocoolers, all of which are since expired [79], leaving CRISM only able to make near-IR observations. Additionally, being the most thermally sensitive instrument, it was shielded by a cover until the end of ATP. The SHARAD instrument uses its support structure as a radiator for its thermal control system, which includes temperature sensors and its own heaters in order to keep its electronics in the operating temperature range [95]. The HiRISE relies on black Kapton MLI covering the outside of its heat-sensitive M55J graphite epoxy baffle. CTX is also covered by MLI, while MARCI relies on an aluminum radiator [19].

5.2.2 Solar Arrays

		Thickness (m)	
Layer	**Material**	**Inboard Panel (T-0190)**	**Outboard Panel (T-0110, T-0309)**
1	Kapton	7.6×10^{-5}	7.6×10^{-5}
2	M55J/RS-3	5.0×10^{-4}	2.5×10^{-4}
3	Al HC 1.6 PCF	2.9×10^{-2}	2.26×10^{-2}
4	M55J/RS-3	5.0×10^{-4}	2.5×10^{-4}
5	Kapton	5.0×10^{-5}	5.0×10^{-5}

Table 17: Material and thickness of solar array layers

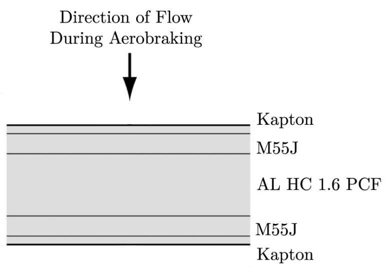


Figure 10: Cross-section [53].

The two solar panels have a total surface area of 20.5 m^2 housing photovoltaic cells, each equipped with a two-axis gimbal system. The solar arrays function as primary drag surfaces during the aerobraking phase of the mission, which results in very high aerodynamic heat loads. The solar array is thus reported as the highest-risk component of MRO during aerobraking [19]. The MRO solar arrays are composed of a multilayered structure, as shown in Figure 10, designed to withstand the thermal stresses, characterized by the materials and thicknesses shown in Table 17.

5.2.3 Batteries

The main power batteries of MRO are two Eagle-Picher 50 Ah, Common Pressure Vessel (CPV) Ni-H_2 batteries, composed of 11 cells each [97]. The discharge efficiency and supplied voltage of Ni-H_2 batteries are highly temperature-dependent, tapering off with increasing temperature [93]. The charge rates are nearly unaffected by temperature within the operating range [93]. The Ni-H_2 batteries show excellent performance at low temperatures (of about -10 °C), and have been demonstrated to survive in fully frozen spacecraft [104].

5.2.4 High Gain Antenna

The 3-meters HGA acts as one of the primary drag surfaces during aerobraking, in addition to its main function of serving the telecommunication goals of the mission. To protect the HGA during aerobraking and orbital temperature swings without hindering communications, it is covered with germanium-coated Kapton [102]. Germanium is preferable in covers for such antennas, as it does not attenuate radio waves. Dedicated heaters keep the Germanium Kapton shielded gimbal mechanism within its operating temperature range [16].

5.2.5 Propulsion Subsystem

The propulsion subsystem of MRO is integrated along the central bus of the spacecraft, with the main thruster assembly extending from aft side. Hydrazine fuel and helium pressurant both exhibit significant sensitivity to temperature variations, prompting the use of a thermally controlled design that leverages both passive insulation layers and heaters to regulate conditions within the propulsion module. Particularly, the propellant lines and tank need to be kept warm to maintain hydrazine above its freezing point and ensure proper thruster operation. The likely solution to meet said requirements (as it is not publicly available) is heaters installed on each tank and line to hold hydrazine above freezing temperature and MLI blankets applied to the tank surfaces to minimize radiative exchange with the surrounding subsystems, as has been the case in some similar missions [90].

5.3 Thermal Fluxes and Phases Correlation

MRO is subjected to various heat sources and sinks throughout its mission. These are summarized in the following table.

	Launch	**Cruise**	**MOI**	**Aerobraking**	**PSP**	**Safe Mode**
Heat Sources						
Solar flux	•	•	•	•	•	•
IR from Earth	•					
IR from Mars			•	•	•	•
Albedo	•*		•	•	•	•
Atmospheric drag				•	•	
Internally generated	•	•	•	•	•	∼
Heat Sinks						
Deep space	•	•	•	•	•	•

Table 18: Presence of heat sources and sinks across different MRO mission phases.
*: From Earth and Moon. ∼: Low to none.

5.4 Reverse Sizing

This section aims to provide a rough order of magnitude thermal analysis of the MRO mission. To do so, the entire S/C is modeled as a single thermal node in steady-state conditions. This simplified time-invariant model does allow the inclusion of a detailed MLI model. It is instead assumed that the entire S/C surface is covered with *"Gold" Kapton*, with known emissivity ε_{SC} and absorptivity α_{SC} values [102].

The assumption of thermal steady-state implies that the total heat entering the S/C equals the amount radiated away. Under this condition, the only unknown in the power balance is the spacecraft temperature, T_{SC}, which is elaborated on in the following sections.
As a starting point, a safety temperature range is derived from Table 16. The safety limits used to define $T_{\max}$ and $T_{\min}$ are based on the average of the maximum and minimum operational temperatures, excluding the extreme outliers with dedicated thermal controls, in this case, CRISM. This approach helps avoid over-dimensioning the radiator area or the required heating power, which would result from setting overly conservative boundaries. In practice, any individual component or subsystem with operational ranges that are outside of this range are managed using dedicated thermal control systems. To simplify the calculation of cross-sectional areas A_{cross}, the spacecraft is approximated as a thermally equivalent sphere. Subtracting solar panels, antennas, and other external devices, the main body of MRO has a surface area of $A_{\text{SC}} = 24.486\,\text{m}^2$, from which $r_{\textbf{sphere}}$ is computed.

$T_{\min}$ w/ margins [°C]	$T_{\max}$ w/ margins [°C]	$r_{\text{sphere}}[\text{m}^2]$	ε_{SC}	α_{SC}	ε_{rad}
-33 (-48 + 15)	62 (77 - 15)	1.396	0.73	0.41	0.85

Table 19: Sizing parameters.

The thermal inputs considered in this analysis are: solar radiation, albedo power, infrared radiation emitted by Mars, aerodynamic drag heating, and internal power dissipation from active subsystems. The heat generated by air drag power comes from the friction between the external surfaces of the S/C and the gas of the atmosphere around it, which in turn is dependent upon the velocity of S/C and the density of the atmosphere, as can be inferred from the corresponding equation below [19].

$$\text{Atmospheric Drag Heating:} \qquad q_{\text{drag}} = \frac{1}{2} \cdot \rho_{\text{atm}} \cdot V^3 \cdot C_{H,\text{ incident}}, \qquad Q_{\text{drag}} = A^{\text{drag}}_{\text{cross}}\, q_{\text{drag}}$$

The coefficient of convective heat transfer due to aerodynamic friction $C_{H,\text{ incident}} = 0.835\,\text{m}^2$ is taken from the paper by J. H. Prince et al. [92], wherein the real heat flux during the aerobraking maneuver of MRO is measured. The density of Mars atmosphere as a function of the S/C altitude is then calculated using a Mars atmosphere model [26]. The $A^{\text{drag}}_{\text{cross}}$ is considered to be equal to the $A^{\text{Sun}}_{\text{cross}}$ because both phenomena can be seen as having an effective area equal to the circle with a radius equal to the one of the equivalent sphere.
The power due to internal heat generated by the spacecraft subsystems is estimated from their respective electrical power consumptions, evaluating independently their electrical-to-thermal efficiencies.
Every object with a temperature above 0 K emits radiation, mostly concentrated in the IR spectrum. The MRO is no exception, as it constantly radiates heat into the colder deep space ($T_{\text{space}} = 3\,\text{K}$). The associated heat flux is directed outward, opposite the thermal influx previously described. It can be derived from the Stefan-Boltzmann equation.
The choices of hot and cold cases are described in the next section, where the heating powers inside Table 20 will be analyzed properly.

Phase	$Q_{\mathbf{Sun}}$	$Q_{\mathbf{ir}}$	$Q_{\mathbf{albedo}}$	$Q_{\mathbf{drag}}$	$Q_{\mathbf{int}}$	$Q_{\mathbf{tot}}$
Hot Case	1485.532	514.387	417.940	9193.200	280.930	1.189e+04
Cold Case	-	384.9	-	0.02	93.9	478.9

Table 20: Heating powers of hot and cold cases (in Watts)

5.4.1 Hot Case

Across the MRO mission, three phases were considered as potential candidates for the occurrence of peak total power due to heat affecting the S/C: LEOP, ATP, and PSP. In the LEOP phase, the S/C is at its closest point to the Sun, therefore Q_{Sun} reaches its maximum value during this phase. However, both Q_{int} and Q_{drag} remain low. Nevertheless, a breakdown of LEOP as the hot case is outside scope of this analysis, as the S/C is still attached to the launcher in parts of this phase. As such, a far more complex thermal analysis would be required. During the PSP, all instruments on board are operational. The spacecraft's proximity to Mars leads to an increase in both Q_{albedo} and Q_{IR}. Since the objective is to identify the worst-case (i.e. peak temperature) scenario, Mars was assumed to be at its perihelion and the S/C at its pericenter around Mars respectively. In the case of the aerobraking phase, the Mars's average distance from Sun was used (due to the lengthy duration of this phase), at the pericenter of the spacecraft's orbit. Despite most instruments being turned off, the high value of Q_{drag} due to the very low altitude (as low as 111 km) and the high relative velocity, this phase is the one in which the highest thermal loads are expected. As such, it is determined as the hot case for this analysis, as can be seen in Table 20. The results for the hot case scenario are as follows:

$T_{\text{max, SC}}$ [°C]	$A_{\text{rad, body}}$ [m^2]	$A_{\text{rad, deployable}}$ [m^2]
96.9	45.2	6.4

Table 21: Hot case results.

5.4.2 Cold Case

Several alternatives were considered for the cold case scenario. The selected condition corresponds to the simultaneous occurrence of a solar conjunction and an eclipse. This configuration occurs multiple times during the mission. During this event, the spacecraft is occluded from the Sun by Mars, and from Earth by the Sun, in addition to being at its maximum distance from the Sun. The eclipse results in the absence of solar radiation and albedo in the thermal power balance. On the other hand, the solar conjunction leads to a communication blackout with Earth, which nullifies the power generation associated with the antennas. As for the spacecraft's internally dissipated power, it was considered to be composed of the standby power consumption of the instruments, since most of them cannot operate during an eclipse. Similarly for the sensors, batteries and core electronics. Another important assumption made, consistent with the previous case, is the use of a steady-state solution. This assumption is particularly relevant in the cold case. During the PSP, the eclipse on Mars lasts about 40 minutes. This short duration implies that the spacecraft does not have enough time to cool down to the equilibrium temperature calculated in this analysis. An additional aspect to consider is the behavior of the deployable radiators, which were proposed as a solution in the previous hot case analysis. These radiators are sized for the hot case, but would still passively radiate heat even when it is not desirable to do so, such as during a cold case scenario. This led to the decision to neglect their contribution during phases where eclipse and solar conjunction occur simultaneously. Although this represents a strong assumption, it can still be considered realistic, given the possibility of implementing such a strategy in practice. Several technical solutions would make this feasible, including the use of thermal louvers or deployable radiators designed to open and close multiple times throughout the mission. In the latter case, the steady-state assumption becomes particularly important once again. The consideration of transients however would significantly complicate the thermal model. The main results for the cold case scenario are as follows:

$T_{\text{min, SC}}$ [°C]	Q_{heaters} w/ margins [W]
-111.5	2321

Table 22: Cold case results.

5.4.3 Results Analysis

Table 21 and Table 22 show the main results of the mono-nodal thermal analysis of the MRO's TCS. These results cannot be considered representative of the actual system behavior, furthermore, they exceed the operational limit set at the beginning of the analysis.
In the hot case, the equilibrium temperature reached without the use of radiators is clearly very high. However, this is somewhat expected, as it depends solely on the thermal properties of the spacecraft's external coating and covers, which may not be sufficient in such a harsh environment. The issue becomes apparent when calculating the required radiator surfaces. The estimated area for body-mounted radiators exceeds the usable surface area available on the spacecraft. This discrepancy may be due to an incorrect estimation of the equivalent sphere's surface, derived from the real spacecraft dimensions. In particular, the contribution of the HGA and solar panels, which represent a sizable portion of the spacecraft's total surface, was not included. When deployable radiators are considered instead, the required surface area is still excessive, rendering this solution unfeasible as well. A way to improve these results is increasing radiator emissivity or adjusting the equivalent sphere's coating, either by raising its emissivity or lowering its absorptivity.
Similarly, the cold case results in an equilibrium temperature that is excessively low, below the minimum allowable limit. To mitigate this, heaters were introduced. However, Table 22 shows that the required heater power (already including margin) is too high, making the solution unfeasible. A potential improvement here would be to reduce the spacecraft's emissivity, though this would contradict the recommendations made for the hot case. Another possible refinement would be to model the MLI layers around the spacecraft more accurately, which would improve its thermal insulation.
In both cases, the mono-nodal analysis proves to be insufficient and unreliable, especially considering that the allowed temperature ranges were already defined to be quite broad. A multi-nodal thermal analysis, focusing on the solar panels and HGA (critical in the hot case) or on the scientific instruments (important in the cold case), would likely yield more accurate and realistic results.

5.4.4 Mass and Power Budget

A more detailed TCS modeling is necessary to derive realistic estimates for mass and power budgets. The reverse sizing process has thus far involved only a preliminary system-level analysis, which does not provide sufficient detail to accurately assess mass and power requirements. Based on the values obtained from the reverse sizing and various reference data [87, 13], such as power density and component density, the estimated mass and power budgets (with a margin of 20%), are as follows:

Mass	45.98 kg
Power	2321 W

Table 23: Mass and power budget.

6 Electric Power Subsystem

The Electric Power Subsystem (EPS) generates, stores, distributes and controls electrical power within MRO. The EPS is designed around a central bus, which operates at a voltage of 28±4 V [21]. This subsystem supplies a continuous source of power to the various electrical loads of the spacecraft during all phases and modes. The following paragraphs aim at providing a complete overview of the EPS, outlining its main features and explaining the rationale behind the solutions adopted.

6.1 Architecture and rationale

The power is generated using two solar arrays and two nickel-hydrogen batteries. The solar panels are mounted on opposite sides of the orbiter with a gimbaled connection, allowing continuous Sun tracking with low aspect angles. The battery supplies the required power during the modes in which solar power generation is not possible, even in case of failure, thanks to the second redundant battery. Figure 11 provides a comprehensive overview of the overall logic and functional architecture of the EPS. It illustrates the main components involved, their interconnections, and the flow of electrical power from generation to distribution and storage. Each of these elements will be analyzed in detail in the following sections, highlighting both their functional roles and design considerations.

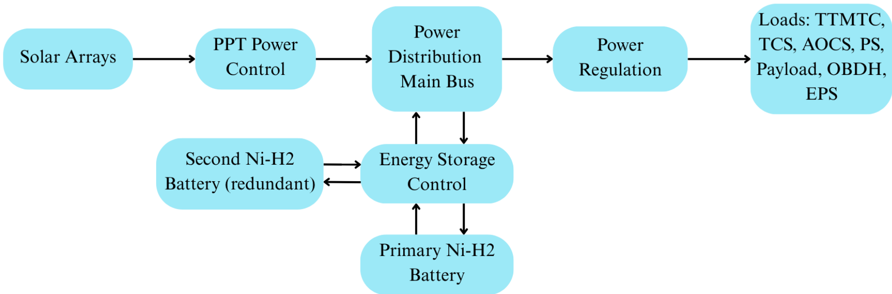


Figure 11: EPS block scheme.

6.1.1 Solar Arrays

Solar panels are lighter and less complex than other energy sources, such as radioisotope thermoelectric generators (RTGs), making solar arrays more suitable for missions wherein spacecraft mass is a critical factor. Thus, as its primary means of power generation, MRO employs two solar arrays placed at either side of the main body. These solar arrays are in a folded configuration at launch and are deployed during checkout (as shown in Figure 12) and remain deployed throughout the entire mission. The selection of solar arrays is also driven by aerobraking maneuver, providing a large area for drag interaction with the Martian atmosphere, saving propellant mass.

Each array is 5.35 m long and 2.53 m wide, with a total effective area of about 10 m^2. On the front side of each array, 9.5 m^2 of surface is covered with 3744 individual photovoltaic cells, which as an array produce 32 V. The solar arrays are capable of producing over 6000 W at Earth and around 2000 W at Mars (where the sunshine is weaker) at the start of the PSP. At the end of the PSP, and at the Mars aphelion, the solar arrays are capable of producing 600 W. The latter value still allows the simultaneous operation of S/C instruments and other subsystems [66, 78, 113]. Other solar array specs are not publicly available, however, given the technological heritage at the time of development and that the solar cells are triple junction with $\varepsilon_{\mathrm{SA}} = 0.26$, they are likely GaInP/GaAs/Ge cells [10]. However, a 6% reduction of the efficiency is estimated by the end of the mission due to degradation [49]. The solar panels have a composite structure, composed of the materials shown in Figure 13. The inner and outer parts of the solar panel, alternatively referred to as *inboard* and *outboard* are made of different thicknesses of said materials, due to their differing thermal and structural requirements [53].

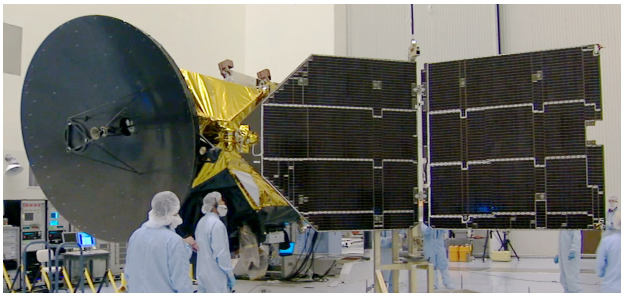


Figure 12: MRO's left solar array, semi-folded during a deployment test [86].

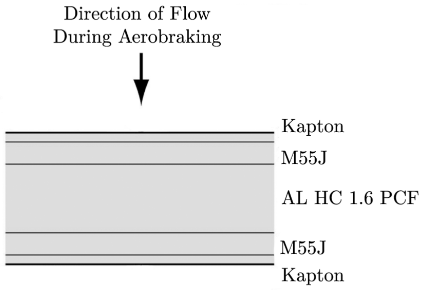


Figure 13: Cross section of the solar panel structure, excluding solar cells and connections [53].

The two-axis gimbal mechanisms of the solar arrays provide increased Sun-facing capabilities (as shown in Figure 14), reducing the average solar incidence angle drastically throughout the mission. While the solar arrays are designed to always point towards the Sun, there are cases wherein this is not possible and thus might pose an undervoltage risk. For example, on November 13th, 2017, during an OTM, the solar arrays were pointed away from the Sun during a sunlit period and MRO subsequently entered eclipse, thereafter battery voltages dropped below critical levels, causing a full safing of the spacecraft [9]. An additional risk caused by this added articulation is collision of the solar arrays with HGA or with the body of the MRO itself, the latter of which took place in the mission due to a mismanagement of the operationally-defined *keep-out zones* meant to prevent self-collision [70].

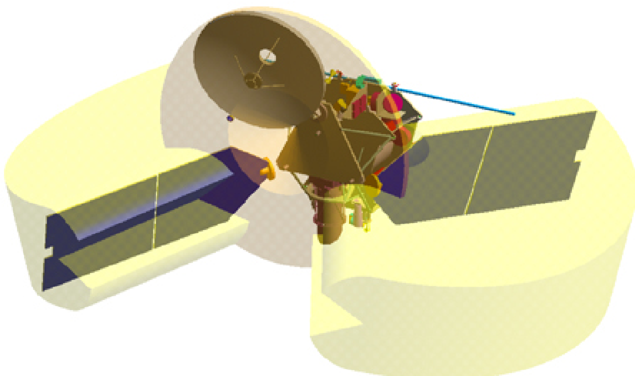


Figure 14: Geometry of MRO two-axis gimbal range of motion [70].

6.1.2 Batteries

MRO's orbit around Mars results in regular transitions between sunlight and shadow, necessitating a power system capable of enduring numerous charge-discharge cycles. The selection and design of batteries are driven by two important aspects, assuring the S/C survivability during Mars eclipses and hold out a high number of cycles, during the primary science and relay phase. The main power batteries of MRO are two Eagle-Picher 50 Ampere-hour, Common Pressure Vessel (CPV) Ni-H_2 batteries, composed of 11 cells each [97]. Each battery is capable of providing 1600 W for one hour at 32 V, even though the nominal voltage is typically set at 28 V. If the voltage drops below 20 V, the on board computer is not able to work and the spacecraft switches to safe mode [78]. The choice of the Ni-H_2 batteries is driven by their long heritage in the space industry, offering reliability and high performances. This technology had accumulated over 30 years of research at the time, having amassed experience in the manufacturing of hydrogen electrode, separator material, electrolyte, overall battery design and life cycle tests [39]. Furthermore, these batteries offer high resistance against radiation and a wide temperature range of operability. The orbit of MRO with a period of around 2 hours results in many cycles over the mission lifetime, necessitating a low depth of discharge (DoD) for the batteries.

After MRO experienced several safings due to low voltage, moving the nominal orbit closer to the terminator (i.e. a later LST) was considered, however, this was evaluated to yield unsatisfactory science imaging results [9]. The MRO team has instead opted to initiate a process of battery reconditioning, wherein the battery and instrument parameters relating to power usage have been tweaked multiple times. Given the extended lifetime of the mission (as of the time of this work, MRO has completed over eighty-nine thousand orbits [84]), it is likely that the DoD has been reduced throughout the various battery management updates, such as the aforementioned battery reconditioning campaign, completed on June 10th, 2019 [9].

6.1.3 Power Control and Distribution Unit

The EPS includes a set of components dedicated to power distribution and control. The regulation is needed because of different aspects, such as load requirements, battery charge control, component degradation, and operational conditions diversity. There are no public sources dedicated to the harness and control logic of this subsystem, however, some considerations can be made based on typical standardized recommendations and previous missions. It is expected to have a centralized power distribution, with cables, fault protection and command decoders. The spacecraft bus voltage is assumed to be regulated at 28 V, consistent with NASA standard electrical system requirements. The EPS can be expected to operate on a Peak Power Tracking (PPT) logic, as it allows for more precise voltage control, which is critical due to the fluctuations in generated power throughout the sunlight-eclipse cycles [10, 27, 49].

6.2 Power Demands Across Mission Phases

Subsystem	**Power Budget [W]**						
	LEOP	**Cruise**	**MOI**	**ATP**	**PSP (Sunlight)**	**(Eclipse)**	**Safe**
TTMTC [40]	135	145.5	155	155	242	71	66
Propulsion [2, 58, 59]	-	122.27	122.27	62.87	62.87	62.87	13.72
AOCS [36, 35, 89, 52, 40]	58.32	63.92	30.92	30.92	120.92	63.6	40.32
EPS [27]	63.5	198.5	198.5	198.5	198.5	300	198.5
TCS [80]	30	250	50	280	30	300	30
OBDH [63]	10	10	10	10	10	10	10
Payload [8, 9, 16, 65, 113]	0	16	0	34	229.4	34	0
Total Power	296.8	790.2	566.7	737.3	893.7	841.5	358.5
Total (w/ margins)	356.2	948.2	680.0	884.7	1072.4	1009.8	430.2

Table 24: Subsystem power demand by mission phase for Mars Reconnaissance Orbiter.

Table 24 reports the average power consumption of the main subsystems across different mission phases or operational modes. It is evident that the power demand varies significantly depending on the spacecraft's operating conditions and requirements. Specifically, the **TTMTC** estimation is based on literature sources, with certain assumptions applied. During the cruise phase, for example, the required power fluctuates considerably due to instrumentation testing, trajectory correction maneuvers, and the increasing Earth-spacecraft distance. Therefore, an average between the minimum and maximum demand was considered. The HGA input power was estimated close to its maximum during the PSP phase, due to the long downlink distance and high data volume, while a lower value was assumed for the earlier phases. The power breakdown of each TTMTC component is provided in the dedicated report. The **AOCS** demand generally remains well below 100 W, except during PSP, due to the high pointing accuracy and frequent attitude maneuvers required for target observations. During MOI and ATP, power consumption is low, as thrusters are used instead of reaction wheels to provide the faster control response needed in those phases. In safe mode, the AOCS relies solely on the MIMU, sun sensors and low reaction wheel control torque. The values of **TCS** are derived from data provided by NASA where a maximum heating power of 300 W is reported [80]. Then, considering on a case-by-case basis the actual heat loads from the various thermal sources, the corresponding electrical power requirements for each mission phase or operational mode are qualitatively assessed. In all phases a minimum of 30 W is imposed to consider the constant heating of the fuel pipes and tanks. For the **Payload** power, all testing and calibration activities are neglected due to their short duration. Therefore, non-zero power values appear only in PSP (full activation of payloads), during eclipse (stand-by mode of instruments), in cruise (ONC) and during ATP (instruments are switched on and in stand-by). **Propulsion** system is characterized by three different types of thrusters. Each of them is used for particular maneuvers during the mission, so different power requirements are demanded in each phase, according to values reported in datasheets [2, 58, 59]. The **On Board Data Handling (OBDH)** subsystem's power requirement is assumed to be equal to its On Board Computer (OBC), as the solid state recorder's power requirement is assumed to be low and is not publicly disclosed [63, 68]. The electric power profile of the control units of the **EPS** subsystem are not available, so the power values required by the battery control unit and that of the solar arrays, are estimated based on information obtained from other missions by the literature [27], therefore the power budget might be oversized with respect to reality.

6.3 Reverse Sizing

While MRO has far surpassed its projected lifetime, it was originally designed to carry out its primary mission until December 31, 2010, or until about 5.4 years after its launch [30].

Using MRO's SPICE data [84] and General Mission Analysis Tool (GMAT), the average eclipse duration (T_e, including penumbrae) over 1000 eclipse events is found to be around 37.1 minutes (± 47 seconds), whereas the average orbit time over three months is 111.9 minutes, leading to an average sunlit duration (T_L) per orbit of 74.8 minutes.

6.3.1 Solar Array Sizing

EoL energy capacity [Wh]	Mass [kg]	Volume [m^3]	Area [m^2]	Cell count	Series $\times$ Parallel
2174.21	96.49	1.1317	21.09	7645	13×590

Table 25: Solar array sizing results

The specific power of GaInP/GaAs/Ge cells is stated as 80 W/kg and their efficiency as $\varepsilon_{\text{SA}} = 0.26$ [10]. Power at the beginning of life, P_{BoL}, is calculated using the solar irradiance at Mars orbit, as launch and cruise phases constitute a minority of the total lifetime. Line efficiencies corresponding to sunlit and eclipsed durations are $X_S = 0.80$ and $X_e = 0.60$ respectively [10]. Yearly degradation factor and inherent degradation factors are assumed dpy $= 0.03$ and $I_d = 0.8$ respectively [10]. Given the large range of movement and the two-axis freedom of motion of the solar wings, it is assumed that the average solar incidence angle over an orbit is 2°. An individual cell is assumed a fairly standard size of 27.5 cm^2. From literature, the end-of-life (EoL) single cell voltage is around 2.3 V for similarly sized 3J solar cells [24, 10]. The solar irradiance on Mars is $P_{0,\,\text{Mars}} = 586.2\,\text{W/m}^2$ [75]. The design lifetime of around 5.4 years is considered. The mass per unit is obtained using the known thicknesses and the densities of the aforementioned constituent materials of the solar panels, in addition to the masses of individual cells (assumed 85 mg). As mentioned before, the inboard has significantly higher mass per unit than the outboard, at $\sigma_{\text{inboard}} = 2.83\,\text{kg/m}^2$ and $\sigma_{\text{outboard}} = 1.71\,\text{kg/m}^2$ respectively, without the solar cells. While exact dimensions are not available, and the outboards are slightly larger due to their lack of tapered corners, for the sake of simplicity, it is assumed that the inboard and the outboard are approximately of the same area and house an equal number of cells. As such, the total mass is:

$$M_{\text{SA}} = 2(\sigma_{\text{inboard}} \times A_{\text{inboard}}) + 2(\sigma_{\text{outboard}} \times A_{\text{outboard}}) + M_{\text{cells}}$$

Using the same thickness values that were used for the mass calculation, with the addition of the assumed cell thickness of 150 µm, the volume of the arrays are obtained.

6.3.2 Battery Sizing

Batteries are the secondary source of power on MRO. The S/C is supplied with two Ni-H_2 battery packs. Each of them is composed of 11 cells in series [97]. Each cell has a voltage of 2.5 V in Direct Current (DC) [101]. The reported DoD is around 40% [78], which is confirmed by the graphical relations in the literature considering the design lifetime of the mission [108]. EoL battery efficiency η and packing efficiency loss factor μ are assumed to be 0.75 and 0.8 respectively, as typical values. Typical values of specific energy and energy density for Ni-H_2 batteries are respectively 55 W h/kg [112] and 50 Wh/l [10]. Once all the properties of the battery packs have been defined, knowing the eclipse duration and having obtained the required power in that period from Table 24, the main results of the sizing procedure, summarized in Table 26, can be computed. The computed results below are relative to a single pack of batteries, and a 20% margin is included on mass and volume:

Energy capacity [Wh]	Electric charge [Ah]	Mass [kg]	Volume [m^3]
1848	67.2	40.32	0.0444

Table 26: Single Battery Pack Sizing Results

6.3.3 Results Analysis

The cell count for the solar arrays is overestimated by a factor of 2.4%. However, changing the efficiency ε_{SA} value from 0.26 to 0.266 yields an exact match with the real cell count and a close agreement with the real populated area. The latter value is in fact slightly closer in line with the high-performing solar cells of the time [62]. This hints at the possibility that the reported solar cell efficiency is that of the EoL. Secondarily, by visually inspecting images of the solar arrays, it can be seen that they are made up of strings of 18 cells, likely in series. Using the reported solar array supply voltage of 32 V [78] instead of the bus voltage of 28 V, the series voltage of 18×2.3 is still in excess. This may have been a measure against the voltage variations of the cells with temperature [69], combined with the additional voltage that would be required to account for inherent degradation at EoL. Additionally, if a maneuver needs to be executed during a satellite day (particularly in PSP), the battery recharge time will be drastically reduced, requiring higher voltages to compensate. The calculated mass of a singular solar array is 48.25 kg, closely in agreement with the real mass of 48.3 kg [19].

Regarding the battery sizing, the results obtained can be considered reasonable in terms of both mass and volume. The only results that can be directly compared with real data from the literature are the number of cells in series and the battery charge. The number of cells in series computed during the sizing matches the actual value, which suggests that the input data used in the sizing process—such as the total system voltage and the single cell voltage—are likely accurate [97]. On the other hand, the battery pack charge obtained from the sizing is higher than the value reported in the literature [97]. This difference is linked to an overestimation of the energy capacity, which is likely due to inaccuracies in defining the battery efficiency and packing efficiency loss factor values. Nevertheless, considering that this is a preliminary analysis, the battery sizing can be considered sufficiently comprehensive.

6.3.4 Mass, Volume, Power and Data Budget

The mass and volume budget are reported below:

Component	**Mass** [kg]	Volume [m^3]
Solar arrays	96.49 (2 × 48.25)	1.1317 (2 × 0.5658)
Batteries	80.64 (2 × 40.32)	0.0889 (2 × 0.0444)
Total	177.13	1.2206
Total (w/ 20% margins)	212.56	1.4647
Total margined with cables	255.07	1.7576

Table 27: EPS mass and volume budget.

The most common power demand from the EPS is estimated at 198.5 W, while the peak power demand is around 300 W which only occurs during eclipse [27]. The MRO query servers operate at a sampling rate of 0.2 Hz for delta - Differential One-way Range (ΔDOR) measurements [96]. Total data rate can be estimated assuming the same sampling rate for telemetry monitoring, with telemetry data containing 10 parameters (such as battery output voltage, solar array output voltage, main bus voltage etc.) averaging roughly 2 bytes each, as a mix of binary and single-precision float values. The resulting estimate is 345.6 kilobytes per day.

7 On Board Data Handling Subsystem

The OBDH is one of the most important subsystems of the MRO, as the mission is characterized by its unprecedented data acquisition rates as well as the relay function of the S/C as part of the DSN. The OBDH subsystem is the core of the S/C's avionics, with the responsibility of connecting the incoming datastream of the S/C instruments with the TTMTC to transmit data to ground stations. When the downlink is not available, the OBDH subsystem needs to be able to locally store the data until the next scheduled downlink pass. The subsystem thus receives telecommands, schedules their processing, gathers and packages telemetry, and maintains time synchronization across all subsystems.

7.1 Architecture and Rationale

The architectural design of MRO's OBDH subsystem adopts a centralized philosophy, with the central OBC governing data handling tasks, and a secondary cold-redundant unit standing by to ensure continuous availability in the event of faults. This architectural simplification helps reducing system complexity and eases integration efforts across the spacecraft's hardware and software layers. Redundancy is applied not only at the processor level but extends to data buses and critical memory interfaces through a cross-strapped configuration. Such a choice ensures that, even under hardware failure conditions, system continuity and responsiveness are preserved [31]. Together, Figure 15 and Figure 16 illustrate both the conceptual and technical layers of MRO's OBDH subsystem. The first shows the cyclic model of science planning and execution, while the second details the encapsulation, storage, transmission, extraction, and archival of scientific data. These images underscore the mission's commitment to data integrity, responsiveness, and traceability throughout its entire lifecycle.

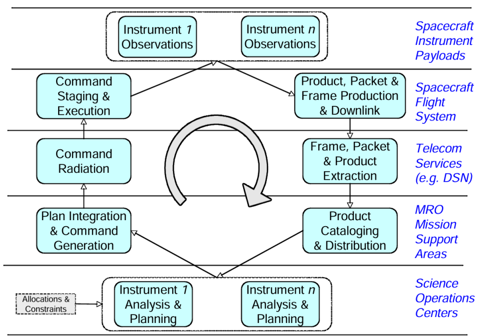


Figure 15: MRO command flow cycle [33]

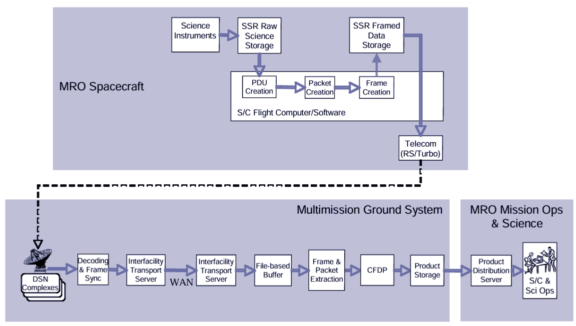


Figure 16: Telemetry data flow [33]

7.1.1 Software

The OBDH software of the MRO plays a central role in managing the spacecraft's high-volume science telemetry. At the software level, the OBDH system adheres to the Consultative Committee for Space Data Systems (CCSDS) standards, structuring its data processing via a layered protocol stack. This begins at the level of Packet Telemetry and Packet Telecommand standards, which define how data is organized and interpreted. For framing and segmentation, the acquisition-of-signal transfer frames protocol is employed, supporting the simultaneous handling of multiple data streams while maintaining frame synchronization and minimizing transmission error impacts. To ensure end-to-end delivery of science products, the CCSDS File Delivery Protocol (CFDP) protocol is applied in unacknowledged mode, a choice that significantly reduces overhead and latency, particularly in high-bandwidth applications [17]. Further extending the software's robustness, the OBDH incorporates mechanisms such as the Encoded Event Data Accountability Table (EEDAT), enabling detailed tracking of data products through all stages, from onboard acquisition to post-downlink processing. The File Delivery Manager

(FDM) complements this functionality by managing data packaging, handling partial retransmissions, and maintaining temporal coherence across fragmented datasets. Fault detection and response are tightly interlinked with these tools; health monitoring routines run persistently in the background, monitoring telemetry metrics and responding autonomously when anomalies or inconsistencies are detected. This high degree of autonomy is vital, especially during intervals of delayed ground contact [68]. Indeed, the on-board software does not rely on product-level acknowledgments; instead, it uses a streamlined approach to support unacknowledged CFDP mode, enabling time-critical data handling with minimal protocol overhead. This architecture allows the MRO flight system to autonomously manage, prioritize, and transmit a continuous stream of science data under the constraints of limited bandwidth and variable downlink opportunities [18, 33]

7.1.2 Hardware

Processor The OBC on MRO consists of the X2000 RAD750 processor, a radiation-hardened evolution of the commercial PowerPC 750 architecture, operating at 133 MHz. This processor, capable of executing up to 48 million instructions per second, supports both spacecraft subsystem control and science payload data handling [66]. The use of a single-board architecture simplifies integration, while also enabling robust fault detection and recovery through layered watchdogs and software health monitoring. The RAD750, chosen for its deep space reliability, resists more than 100 krad of radiation and prevents single event latch-ups, crucial for Mars missions. It runs VxWorks flight software, a modular system managing core functions, payloads, and fault protection [66]. This last module constantly scans telemetry, triggers autonomous recovery when possible, or shifts the spacecraft to safe mode pending ground support vital during long Earth-Mars delays [66, 68]. However, its architecture simplifies the integration of both science and housekeeping functions within a unified software environment. Its robust watchdog infrastructure monitors process health continuously, with layered fail-safes capable of initiating soft resets or even triggering a complete swap to the redundant unit in case critical issues arise. This approach to fault mitigation, relying heavily on autonomous software handling, reflects a deliberate design philosophy prioritizing operational continuity.

Memory and Storage MRO makes use of a Solid-state Data Recorder (SSDR) supplied by the company SEAKR Engineering [74], which has a 160 Gbit data storage capability, and degrades to 100 Gbit at end-of-mission. The SSDR consists of an array of 720 Hitachi synchronous DRAM cards with each 256 Mb memory size [105]. The total capacity is divided into two different areas of memory; one dedicated to the raw science data from the instruments, the other specifically for the acquisition-of-signal framed data awaiting downlink. To ensure the instruments can operate independently from each other, the raw science area is further hard partitioned, ensuring that each instrument has enough space for its raw science data. The writing of the data on the SSDR is done by the instrument software itself, while the reading and processing is dedicated to the S/C's Flight Software (FSW). Only OBC and its software are allowed to frame, manage, and queue telemetry products. The packages are eventually transferred to the SDST for downlink, adhering to a tightly controlled first-in-first-out (FIFO) scheduling logic that guarantees timely and ordered delivery. The other half of the SSDR was first configured as two hard partitions, one for both the X-band and Ka-band downlink. However, since the Ka-band failure, the whole half of the SSDR is set up for the X-band, which includes further smaller soft partitions for each of the instruments. The computer also has available 128 Mb of high-speed, random-access memory [38, 66]
Assuming that the agreement between instrument operators is based on the total data acquisition volume per instrument, the percentage of SSDR would be as described in Table 28. Engineering, Spacecraft Planet Instrument C-matrix Events (SPICE), MCS, tracking (for gravity science) and accelerometer data are not counted because of their small volumes.

Instrument	Allocation	Data Volume [Tbits]
CRISM	30%	7.8
CTX	13%	3.5
HiRISE	35%	9.1
MARCI	7%	1.6
SHARAD	15%	4.0

Table 28: MRO data volume allocations.

Bus Architecture Communications within the spacecraft are managed through a combination of robust and high-speed data buses chosen for their specific roles. The command and telemetry backbone employs the MIL-STD-1553B bus standard, chosen for its reliability, deterministic timing, and dual redundancy capability. This bus operates at 1 Mbps, which is sufficient for system control, mode transitions, and low-volume data streams. All main subsystems and instruments interface with this bus as Remote Terminals (RTs), responding to polling sequences initiated by the OBC acting as the bus controller [94]. However, due to the limited bandwidth of 1553B, science instruments with high data output, such as HiRISE and SHARAD, utilize dedicated high-speed links to transfer large volumes of data directly to the SSDR. These links are likely to be implemented using RS-422 or LVDS serial protocols, which offer better throughput and electrical robustness. This dual-path bus approach allows control traffic to remain isolated from payload data, preventing congestion, and ensuring reliability across mission phases.

Telecom Interface The final stage in the data chain involves transferring the framed science telemetry from SSDR to the SDST, responsible for modulating and transmitting data through the high-gain antenna of the spacecraft. Here, the MIL-STD-1553B bus is again used for control and configuration tasks, while the high-speed telemetry data path is likely managed via a dedicated high-throughput channel. The SDST supports data rates up to 6 Mbps, making it suitable for MRO's ambitious science return volumes [3]. Redundant cross-strapping is also implemented, allowing either of the two OBC units to interface with the SDST and thus maintaining mission continuity in the event of failure on the OBC side.

7.2 Data Budget

MRO's data budget, divided per subsystem, is shown in Table 29. The following results were estimated by comparison with previous missions or by assumptions based on typical values. In particular, the sampling frequencies were estimated based on the changing rate of the measured parameters; data types were chosen according to typical spacecraft telemetry encoding standards; number of values, n, were selected based on the number of components, and their respective outputs [4, 5, 11, 14, 25, 40, 60]. Regarding the P/Ls, the data rate has been estimated based on the total amount of data generated during the PSP phase, as shown in Table 28, and then averaged over the duration of the PSP. The total estimated data rate is 0.409 Mbps, slightly below the typical operational range floor of 0.5 Mbps [40, 40], that's likely due to omitted parameters or underestimated payload contributions. Assuming a 4-hour storage window between downlinks [40], this yields a total data volume between downlink events of 5.892 Gb, increased to 23.568 Gb with a conservative 400% margin. It is important to note that the mass memory capacity of MRO is 160 Gb [105], which comfortably exceeds the estimated requirement. While this may initially suggest an oversized design, it is not the case: the calculated data volume is based on a highly approximate and conservative lower-bound estimate of the actual data rate and there are periods, such as solar conjunctions, when communications with Earth are completely suspended for weeks and all collected data must be stored locally, justifying the large memory size.

Parameter name	**n.**	**data type**	**bit size**	**frequency** [Hz]	**data rate** [bps]
Electric Power Subsystem (EPS)					
Solar array voltage	1	uint16	16	1	16
Solar array current	1	uint16	16	1	16
Battery voltage	1	uint16	16	1	16
Battery current	1	uint16	16	1	16
Bus voltage	1	uint16	16	1	16
Battery temperature	2	sint16	16	0.5	16
On Board Data Handling (OBDH)					
OBC temperature	1	sint16	16	1	16
OBC voltage	1	uint16	16	1	16
OBC current	1	uint16	16	1	16
Watchdog timer	1	uint8	8	0.1	0.8
Boot counter	1	uint8	8	0.1	0.8
Packet count in memory	1	uint16	16	0.01	0.16
Attitude and Orbit Control Subsystem (AOCS)					
Star tracker output	3	float64	64	1	192
Gyro output	3	float32	32	10	960
Sun sensor raw output	24	sint16	16	5	1920
Estimated Sun vector	3	float32	32	2	192
Estimated quaternion	4	float32	32	1	128
ADCS mode	1	uint8	8	0.1	0.8
IMU current	1	sint16	16	1	16
IMU voltage	1	sint16	16	1	16
IMU temperature	1	sint16	16	0.5	8
Propulsion Subsystem (PS)					
Tank pressure	2	uint16	16	0.1	3.2
Tank temperature	2	uint16	16	0.1	3.2
Valve status	4	bool	1	0.1	0.4
Tracking, Telemetry and Telecommand Subsystem (TTMTC)					
Transponder status	2	uint8	8	0.2	3.2
Link error flag	2	bool	1	0.1	0.2
Thermal Control Subsystem (TCS)					
Radiator temperatures	8	uint16	16	0.1	12.8
Heaters status	4	bool	8	0.1	3.2
payload (P/L)					
HiRISE	-	raw	-	-	141946.2
SHARAD	-	raw	-	-	60834.1
CRISM	-	raw	-	-	121668.2
MARCI	-	raw	-	-	28389.2
CTX	-	raw	-	-	52722.9

Table 29: MRO data budget per subsystem.
-: unavailable.

8 Structure & Configuration Subsystems

The structural subsystem includes the support structure, mechanisms, launcher adapter, and moving parts. The structure and configuration of MRO are driven by several factors, the most critical of which is the necessity of a dedicated nadir-facing instrument deck to fulfill the scientific objectives of the mission. This restriction in turn has driven the design of the structure and the various configurations of MRO throughout the mission. Additionally, the scientific instruments on the nadir deck have to be protected from intense aerodynamic stresses and thermal loads during aerobraking and plume impingement from thrusters during maneuvers, while avoiding direct solar exposure.

The primary structure is the main load-carrying structure, and thus is sized to withstand launch accelerations (-2 to 5g axially, 2g laterally [107]) and dynamic interactions. The primary structure of MRO is composed of deck plates aligned alongside central bus housing the propellant tank, as shown in Figure 17. These plates are referred to by their positions, namely: aft, middle, forward (HGA baseplate), and nadir decks. These decks are joined via support struts and gussets, integrated with the load-bearing central bus, which houses the propellant tank. The gussets stiffen the structure, i.e. raise the mode frequencies above the frequencies observed during launch and aerobraking, specifically, the low-frequency vibration that tends to be the design driver for S/C structure, which constitutes the 0-50 Hz range [107]. The plates are made of aluminum honeycomb structures, and the central bus is made of titanium and reinforced with carbon-fiber reinforced polymer (CFRP) [77], which yields a structure high in mass-to-strength ratio.

The secondary structure of MRO includes deployables and supports for components, such as the solar arrays and SHARAD. The HiRISE is mounted via a rigid support ring connected to the middle deck, while the housing for the instrument itself is graphite [9].

The entirety of the S/C body and the HGA are covered in MLI blankets for thermal insulation, which provide little to no structural support, but add mass and thickness and affect the vibrational response of the S/C. The mass of the structure subsystem is calculated as the remainder when all of the previous subsystems are subtracted from the total mass, which is 352.25 kg, most of which is composed by the titanium tank.

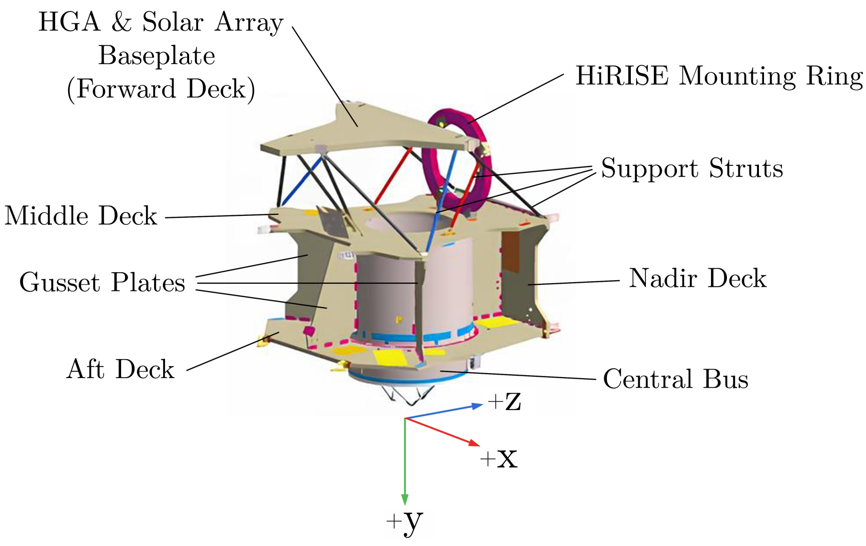


Figure 17: A CAD model of the structure of MRO, adapted from [77].

8.1 Mechanisms and Deployables

MRO employs several mechanisms to enable the deployment and pointing of its components. These include three two-axis gimbals: one for the HGA, providing precise and continuous Earth-pointing capability, and two for the solar arrays, enabling Sun-tracking throughout the mission. All of the gimbals are designed to withstand significant static and vibrational loads during launch, MOI and aerobraking. Specifically, they shall resist and backdrive the reaction torques observed during all phases. For each axis, the gimbals utilize a two-phase brushless DC motor with a direct drive to the wave generator of a size-32 cup-type harmonic gear [12]. The gimbal actuators with these specifications were chosen to —in addition to bearing the launch loads —minimize jitter and achieve smooth rotor velocity [12], both of which are critical for the quality of scientific imaging. Since the gimbals are all single points of failure, the usage of heritage materials and processes were greatly prioritized [12].

In addition, SHARAD, a 10-m dipole antenna made up of two 5-m foldable tubes, is stowed until the transition phase (as shown in Figure 21) and utilized from PSP onward. SHARAD is made using elastic fibers, and thus self-deploys once released via solenoid-controlled actuators [95].

8.2 Configuration Across Phases

To protect the S/C during ascent and partially during LEOP, it is enclosed by a payload fairing (PLF). MRO is launched within the 4.2-m diameter PLF connected to the Centaur (the upper stage of the launch vehicle Atlas V 401 [107]), shown in Figure 18. The folded configuration is therefore driven by the inner dimensions of the PLF.

Figure 18: MRO in its launch configuration, next to the Centaur payload fairing [88].

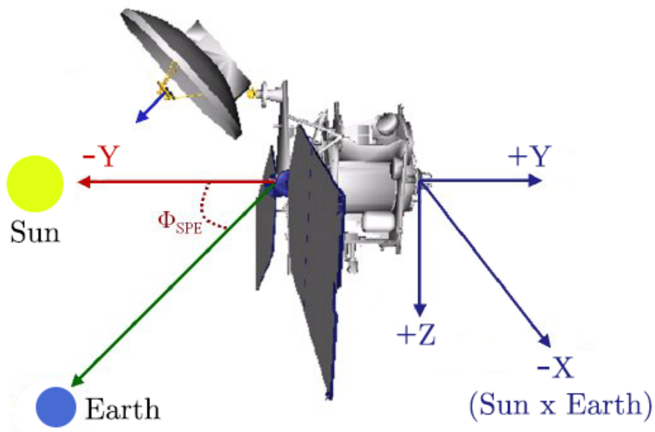


Figure 19: Cruise configuration of MRO, adapted from [110].

MRO is connected to the Centaur via an aluminum 1.2-m diameter payload adapter on the +Y face of the S/C. The payload adapter, positioned at the end point of the propellant tank, transmits the launch loads to MRO, and is jettisoned away along with the payload separation ring using a releasable clamp band after launch, separating with springs [83].

The solar arrays are in a folded position at launch (Figure 18), are deployed during checkout, and assume different configurations depending on mission phase. Specifically, they are in a spread-eagle configuration during aerobraking to maximize drag surface and during cruise (as shown in Figure 19),, in *SHARAD park position* (pointing away from SHARAD) during orbital night and occultations to minimize electromagnetic interactions between the arrays and the instrument [22], and are nominally Sun-tracking during orbital days in PSP and EM, as shown in Figure 20.

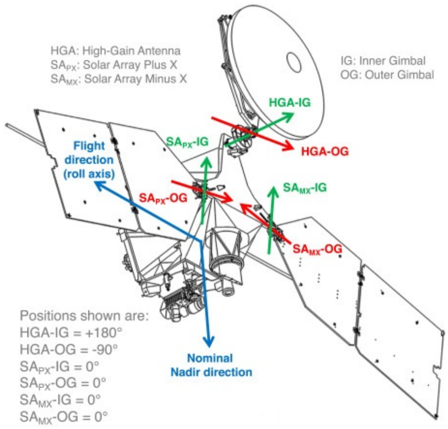


Figure 20: Nominal configuration of MRO during PSP and EM [22].

8.3 External Components and Distribution of Appendages

The solar arrays of MRO are mounted at the +X and -X edges of the forward deck. This results in a balanced mass distribution and symmetric aerodynamic loading, which help with the positioning of the CoM and stability during aerobraking respectively. During nominal operations, owing to both the 3pm LST orbit and the lateral placement of the arrays, self-shadowing is minimal. The solar arrays remain fixed during the cruise phase as the AOCS maneuvers MRO to point the arrays at the Sun, with no self-shadowing of critical systems throughout this phase [110]. This configuration is therefore optimal for ensuring continuous solar exposure over the entire mission.

HGA is mounted on the HGA baseplate (forward deck), and is designed to stow flush against the -Z face of the body at launch, deploying perpendicularly away from the body during the aerobraking phase (in conjunction with the solar arrays) to maximize the drag surface while achieving a stable aerodynamic configuration. In this configuration, the center of pressure is aligned with the CoM along the Y-axis of the craft, and the aft deck nominally faces against the direction of flow.

SHARAD is placed at the +Y edge of the -Z face, which is permanently in shadow during nominal operations as a result of the 3pm LST orbit. In addition to minimizing thermal fluctuations and solar noise, this placement is furthest away from the solar arrays, which are the components that most impede the signal performance of the radar sounder [22].

The scientific instruments (shown in Figure 21), with the exception of SHARAD, are all situated on the nadir deck, pointing towards the +Z direction. The components on the nadir deck are not at risk of damage during the aerobraking, as they are downstream of the shock generated near the aft deck, where the energy content of the flow is lower [28].

The position of the ONC, which is towards the +Y edge of the -Z face (next to SHARAD) is informed by the cruise attitude, as it shall point towards Mars during approach to measure the locations of its satellites and ascertain distance [56]. Another engineering instrument, the UHF antenna is also situated on the nadir deck, which facilitates communication with surface vehicles on Mars. Placement of the UHF antenna in the vicinity of the instruments results in electromagnetic interference between these components, therefore the UHF antenna is activated when the science instruments are not and vice versa [40].

While there is no publicly available information regarding the distribution of the TCS, passive cooling radiators are presumed to be mounted on the aft deck, facing deep space throughout nominal operations.

Star trackers similarly function when facing deep space, and should be kept from direct solar exposure. As can be seen in technical drawings of MRO (e.g. Figure 4), the two redundant star trackers are positioned at the -Z face of the body (zenith direction), and nominally face deep space throughout scientific operations, relieving the spacecraft of the necessity to frequently execute attitude data acquisition maneuvers. Due to its orbit, the star trackers do not face the Sun and can function at any section of the flight.

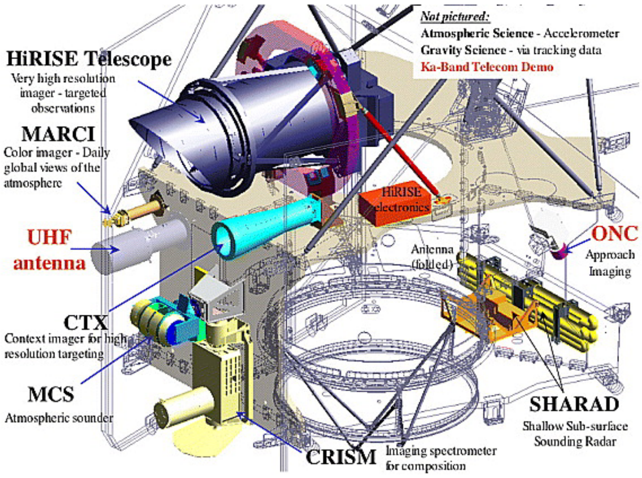


Figure 21: The science and engineering payloads onboard MRO at launch configuration. Middle and nadir deck plates are colored [113].

The six main thrusters are placed axisymmetrically around the centerline of the propellant tank [110], such that the only residual torque is generated is due to mounting error and thrust imbalances. Whereas the secondary thrusters are positioned in a coupled manner on the outer edges of aft and middle decks [110], resulting in a large lever arm and plume flow distant from the appendages.

8.4 Internal Components

The RWA, consisting of four wheels, (three orthogonal for full three-axis control, and one redundant [42]) is likely placed along the body frame axes of the S/C, near the CoM as such a positioning would yield a simpler control law and lower control torques. Wheels generate significant tonal and broadband vibrations due to mass imbalances and manufacturing imperfections, relating to the structural modes of the wheels. As many of the science instruments (specifically HiRISE) have tight jitter requirements [51], a detailed analysis of the wheel disturbances is required to ensure structural stability.

The placement of the propellant tank is constrained by a need to maintain a manageable CoM, as the CoM shifts with propellant consumption. As can be seen in Figure 17, it is axially aligned through the Y-axis of MRO. Using a simplified CAD model (introduced in section 4), it was confirmed that due to its approximately in-line alignment, the propellant tank allows the net thrust vector of the main thrusters to be aligned with the CoM throughout the mission.

The OBDH components and electronic units of the instruments are housed primarily on the middle and aft decks [81]. Having the electronics situated in close proximity to each other minimizes line losses. The batteries are presumably mounted centrally, in a position which allows both reduced harness lengths (thereby lower voltage drops), and harness routing that minimizes inductive coupling with other components, while not increasing control torques drastically.

8.5 End-of-Life Constraints

Any EoL considerations regarding planetary protection provisions have not affected the design of MRO, since all of its components, including the structure subsystem have been analyzed to exceed the temperature limit to be considered sterile upon Mars entry [6].

9 Mass Tables

Subsystem (Dry mass contribution)	**Mass w/o margin [kg]**	**Margin [%]**	**Mass w/ margin [kg]**
STR	268.06	10	294.9
TCS	30.93	10	34.0
TTMTC	61.86	20	74.2
OBDH	134.03	10	147.4
ADCS	92.79	10	102.1
PS	32.58	10	35.8
EPS			
Solar Arrays	96.49	20	115.8
Batteries	80.64	20	96.8
Subtotal	177.13	20	**212.6**
Instruments	113.41	50	170.1
Total dry mass	910.79		1071.1
System margin		20	
Total dry mass w/ margin			1285.4
Propellant	1017.32	5.5	1073.2
Total wet mass			2358.6
Adapter mass	61.8	5	64.9
Total launch mass (w/ adapter)	2423.5	5	2544.7

Table 30: Subsystem mass breakdown as computed through reverse engineering

Subsystem (Dry mass contribution)	**Mass w/o margin [kg]**	**Margin [%]**	**Mass w/ margin [kg]**	**References**
STR	-	-	314.7	[54]
TCS	-	-	20	[20]
TTMTC	-	-	107.7	[40]
OBDH	-	-	32	[20]
ADCS	-	-	101	[20]
PS	-	-	176	[20]
EPS				
Solar Arrays	-	-	96.6	[19]
Batteries	-	-	51	[101]
Subtotal	-	-	148	
Instruments	-	-	132	[38]
Total dry mass	-	-	1031	
System margin			-	
Total dry mass w/ margin			-	
Propellant	-	-	1149	
Total wet mass			2180	
Adapter mass	-	-	61.8	
Total launch mass (w/ adapter)	-	-	2241.8	

Table 31: Subsystem mass breakdown as reported in literature

Table 30 displays the estimated masses of the main subsystems of the MRO mission. Among these data, only those of EPS and PS are computed using a full reverse engineering approach from scratch. For all the other subsystems, the values are evaluated by applying the typical mean mass percentage related to those subsystems, based on other planetary missions found in tables [50], to the real total dry mass of 1031 kg. Then, margins were applied. The margins of the specific subsystem were evaluated one by one, based on the available information on the MRO systems. Among the selection criteria were the TRL that certain components were believed to have, as well as the specific importance of some subsystems in terms of mission success.

Table 31 reports the real values found in literature for all main subsystems. Some of said values, due to the lack of published data, come from a primary estimation made by the MRO's design team ahead of the actual development of the project.

- **STR:** The structural mass estimated through reverse engineering is slightly lower than the actual one, yet both values fall within the typical percentage ranges observed in comparable interplanetary missions. The two values can be considered consistent despite the difference.
- **TCS:** The total mass of the TCS subsystem is relatively aligned across both datasets. A minor difference exists, but it remains within acceptable margins and does not raise concern.
- **TTMTC:** A 20% margin was applied to the estimated TTMTC subsystem mass due to its central role in mission success. The inclusion of both HGA and LGA systems, designed to handle unprecedented data rates, as well as three high-mass TWTAs are elements that surely increase the mass of this subsystem in respect to other missions. Indeed, the real mass value significantly exceeds the estimated one, even after applying the margin.
- **OBDH:** This subsystem has a large mismatch between estimated and actual mass. The discrepancy is likely due to an underestimation during the early design phases of the MRO. Interestingly, the estimated value appears more realistic, because just by considering the internal buses needed to manage the exceptional data throughput required by the mission, the mass already exceeds the real value.
- **AOCS:** For the AOCS subsystem, the estimated and actual values are nearly identical, indicating a good approximation during the preliminary design phase.
- **PS:** The estimated mass of the PS was derived from a reverse engineering approach of tanks and feed lines for both propellant and pressurant. The actual value is significantly higher, mainly due to the presence of additional components such as valves, sensors, redundant lines, and all secondary elements required for proper operation, which were not included in the analysis.
- **EPS:** Similarly to TTMTC, a 20% margin was applied to the EPS estimation. This is due to the highly approximated calculation of both solar array and battery masses, and the complete exclusion of other essential components. The real mass is notably lower than the estimated one, highlighting the limited accuracy of the initial assumption.
- **INSTRUMENTS:** The actual payload mass is obtained by summing up all declared instrument masses. The estimated value, which includes a 50% margin to account for the natural variability between scientific goals and novel instrumentation technologies, turned out to be reasonably close to the real one.

Overall, the comparison between estimated and literature-based values shows a reasonable alignment across most subsystems, especially considering the data availability and the assumptions applied during the estimation process.

The global comparison confirms that the methodology used for mass estimation is suitable for preliminary mission design, providing a balance between conservatism and realism.

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Acronyms

ΔDOR delta - Differential One-way Range. 39

ABM Aerobraking Maneuver. 7, 8

ABX Aerobraking Termination Maneuver. 7

ADE axial displaced elliptical. 14, 15

AM Aerobraking Mode. 23, 26

AMD angular momentum desaturation. 9, 23

AOCS Attitude and Orbit Control Subsystem. 20–22, 24, 26–28, 36, 37, 43, 46, 50

APE Absolute Performance Error. 24

ATP Aerobraking and Transition Phase. 2, 29, 32, 36, 37

BER bit error rate. 18

BPSK Binary phase-shift keying. 16, 18

CAD computer-aided design. 44, 47

CAP Cruise and Approach Phase. 2, 6

CCD Charge-Coupled Device. 21

CCSDS Consultative Committee for Space Data Systems. 40

CFDP CCSDS File Delivery Protocol. 40, 41

CFRP carbon-fiber reinforced polymer. 44

CM Checkout Mode. 22, 26

CoM Center of Mass. 22, 46, 47

CPV Common Pressure Vessel. 35

CRISM Compact Reconnaissance Imaging Spectrometer for Mars. 4–6, 20, 23, 24, 28, 29, 31, 43

CTX Context Camera. 5, 6, 20, 23, 24, 28, 29, 43

DC Direct Current. 21, 38, 45

DL downlink. 16, 18, 19

DLA Declination of Launch Asymptote. 7

DoD depth of discharge. 36, 38

DRA Direct Radiating Array. 14

DSN Deep Space Network. 15–17, 40

ECC Error-correction code. 17

EDL Entry, Descent, and Landing. 3

EEDAT Encoded Event Data Accountability Table. 40

Electra Electra Proximity Link Payload. 3–6, 14, 15, 17

EM Extended Mission. 14, 45, 46

EMI electromagnetic interference. 15

EoL end-of-life. 37, 38, 47

EPS Electric Power Subsystem. 34, 36, 37, 39, 43, 50

FDM File Delivery Manager. 40

FIFO first-in-first-out. 41

FIT Failure in Time. 21

FOV Field-of-view. 4, 20, 21

FSK Frequency-shift keying. 16

FSW Flight Software. 41

GMM Global Mapping Mode. 23, 26

HGA High-Gain Antenna. 4, 14, 15, 17–20, 22–24, 30, 33, 35, 36, 44–46

HiRISE HIgh Resolution Imaging Science Experiment. 4–6, 20, 23–25, 28, 29, 42–44, 47

IMU Inertial Measurement Unit. 27

IR infrared. 28, 29

ITL Integrated Target List. 2

ITU International Telecommunication Union. 18

LEOP Launch and Early Orbit Phase. 2, 16, 17, 32, 36, 45

LGA Low-Gain Antenna. 14–17, 23, 24

LGA1 Low-gain Antenna 1. 23

LST Local Solar Time. 28, 36, 46

LVDS Low-Voltage Differential Signaling. 42

MARCI Mars Color Imager. 4, 6, 23, 24, 28, 29, 43

MCS Mars Climate Sounder. 4–6, 20, 23, 24, 41

MEP Mars Exploration Program. 15

MIMU Miniature Inertial Measurement Unit. 21, 22, 26, 28, 37

MLI Multi-Layer Insulation. 29–31, 33, 44

MOI Mars Orbit Insertion. 2, 4, 6–12, 15, 17, 22–24, 36, 37, 45

MRO Mars Reconnaissance Orbiter. 1–4, 6, 7, 9–11, 14–25, 28–42, 44–47, 50

NEA Noise Equivalent Error. 21

OA Orbit Adjustment. 7, 8

OBC On Board Computer. 37, 40–42

OBDH On Board Data Handling. 36, 37, 40, 43, 47

ONC Optical Navigation Camera. 5, 20, 28, 46

OTM Orbit Trim Maneuver. 8, 35

P/L payload. 1, 3–5, 15, 42, 43

PLF payload fairing. 45

PPT Peak Power Tracking. 36

PS Propulsion Subsystem. 9, 11, 13, 43, 50

PSK Phase-shift keying. 16

PSO Primary Science Orbit. 1, 2, 4, 7

PSP Primary Science Phase. 1–3, 5, 6, 9, 14, 17, 22–26, 32, 34, 36–38, 42, 45, 46

QPSK Quadrature phase-shift keying. 16, 17

RCS Reaction Control System. 20–22, 26, 27

RP Relay Phase. 3, 6

RTs Remote Terminals. 42

RW reaction wheels. 20–22, 25–28

RWA Reaction wheel assembly. 27, 47

S/C spacecraft. 1, 14, 18, 26, 28, 29, 31, 32, 34, 35, 38, 40, 41, 44, 45, 47

SAFE Safe Mode. 23

SCM Sun-point Cruise Mode. 23, 26

SDST Small deep space transponder. 16, 41, 42

SEM Spread-Eagle Mode. 23, 26

SEP Sun-Earth-Probe. 18

SHARAD SHAllow Subsurface RADar. 5, 6, 23, 24, 28, 29, 42–46

SNR signal-to-noise ratio. 17, 18

SPICE Spacecraft Planet Instrument C-matrix Events. 41

SRP solar radiation pressure. 24, 25

SSDR Solid-state Data Recorder. 41, 42

SUR Survey Mode. 23, 26

TCM trajectory correction maneuver. 6–9, 17, 22, 23

TCS Thermal Control Subsystem. 28, 33, 36, 37, 43, 46, 50

TIM Target Investigations Mode. 23, 26

TMTC Telemetry and Telecommand. 17, 19

TRL Technology Readiness Level. 21, 50

TTMTC Tracking, Telemetry and Telecommand Subsystem. 20, 28, 36, 37, 40, 43, 50

TWTA traveling-wave tube amplifier. 14, 15, 19

UHF Ultra High Frequency. 3, 5, 14, 15, 46

UL uplink. 16, 18, 19